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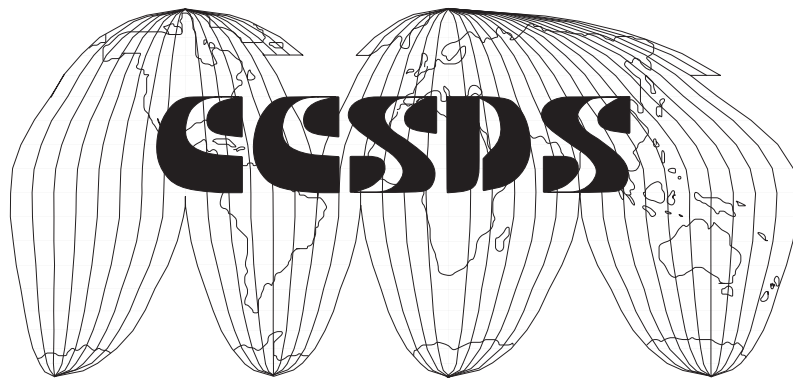
**DRAFT RECOMMENDATION FOR SPACE
DATA SYSTEM STANDARDS**

SPACE LINK EXTENSION FORWARD CLTU SERVICE SPECIFICATION

CCSDS 912.1-R1.99

RED BOOK

February 2000



AUTHORITY

Issue:	Red Book, Issue 1.99
Date:	February 2000
Location:	Pasadena, U.S.A

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National Aeronautics and Space Administration
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This document is a technical **Recommendation** for use in developing ground systems for space missions and has been prepared by the **Consultative Committee for Space Data Systems** (CCSDS). The Space Link Extension Forward CLTU Service referenced herein is intended for missions that are cross-supported between Agencies of the CCSDS.

This **Recommendation** specifies a data service that extends certain of the space-to-ground communications services previously defined by CCSDS (references [2] and [3]) within the framework established by the CCSDS Space Link Extension Reference Model (reference [1]). It allows implementing organizations within each Agency to proceed with the development of compatible derived Standards for the ground systems that are within their cognizance. Derived Agency Standards may implement only a subset of the optional features allowed by the **Recommendation** and may incorporate features not addressed by the **Recommendation**.

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DOCUMENT CONTROL

Document	Title	Date	Status and Substantive Changes
CCSDS-912.1-R-1	Space Link Extension Forward CLTU Service	November 1997	Original Issue
CCSDS-912.1-XX	Space Link Extension Forward CLTU Service	April 1998	Updates from P3 1/98 meeting in Darmstadt and 3/98 working group meeting in Pasadena
CCSDS-912.1-XX	Space Link Extension Forward CLTU Service	September 1998	Aligned with Forward Space Packet for review at 9/98 working group meeting in Oberfaffenhofen.
CCSDS-912.1-R-1.9	Space Link Extension Forward CLTU Service	June 16, 1999	Updates from P3 5/99 meeting in Newport Beach
CCSDS-912.1-R-1.93	Space Link Extension Forward CLTU Service	June 30, 1999	Editorial changes in preparation for submission to CCSDS Secretariat
CCSDS-912.1-R-1.98	Space Link Extension Forward CLTU Service	July 30, 1999	Alignment with CCSDS 912.3-R-1.9
CCSDS-912.1-R-1.99	Space Link Extension Forward CLTU Service	February 2000	Updates from P3 10/99 meeting in Darmstadt

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1 INTRODUCTION

1.1 PURPOSE OF THIS RECOMMENDATION

This Recommendation defines the Command Link Transmission Unit (CLTU) service in conformance with the transfer services specified in reference [1], *Cross Support Reference Model—Part 1: SLE Services*. The Forward CLTU service is an SLE transfer service that enables a mission to send Command Link Transmission Units (CLTUs) to a spacecraft.

1.2 SCOPE

1.2.1 This Recommendation defines, in an abstract manner, the Forward CLTU in terms of:

- a) the operations necessary to provide the transfer service;
- b) the parameter data associated with each operation;
- c) the behaviors that result from the invocation of each operation; and
- d) the relationship between, and the valid sequence of, the operations and resulting behaviors.

1.2.2 It does not specify:

- a) individual implementations or products;
- b) the implementation of entities or interfaces within real systems;
- c) the methods or technologies required to radiate data to a spacecraft and to acquire telemetry frames from the signals received from the spacecraft for extraction of the Operational Control Field;
- d) the methods or technologies required to set up the environment required for communications;
- e) the management activities necessary to schedule, configure, and control the Forward CLTU service.

1.3 APPLICABILITY

1.3.1 APPLICABILITY OF THIS RECOMMENDATION

This Recommendation provides a basis for the development of real systems that implement the Forward CLTU service. Implementation of the Forward CLTU service in a real system additionally requires the availability of a communications service to convey invocations and returns of Forward CLTU service operations between service users and providers. This Recommendation requires that such a communications service ensures that invocations and returns of operations are transferred:

- a) in sequence;
- b) completely and with integrity;

- c) without duplication;
- d) with flow control;
- e) with notification to the application layer in the event that communications between the service user and the service provider are disrupted, possibly resulting in the loss of data.

It is the specific intent of this Recommendation to define the Forward CLTU service in a manner which is independent of any particular communications services, protocols, or technologies.

1.3.2 LIMITS OF APPLICABILITY

This Recommendation specifies the Forward CLTU service that may be provided by an SLE System for inter-Agency cross support. It is neither a specification of, nor a design for, real systems that may be implemented for the control and monitoring of existing or future missions.

1.4 RATIONALE

The goal of this Recommendation is to create a standard for interoperability between the tracking stations and ground data handling systems of various agencies, and the users of forward services.

1.5 DOCUMENT STRUCTURE

1.5.1 DOCUMENT ORGANIZATION

This Recommendation is organized as follows:

- a) Section 1 provides purpose, scope, applicability, and rationale of this Recommendation and lists definitions, nomenclature, conventions, and references used throughout the Recommendation.
- b) Section 2 presents an overview of the Forward CLTU service including a functional description, the service management context, and protocol considerations;
- c) Section 3 specifies the operations of the Forward CLTU service;
- d) Section 4 specifies the dynamic behavior of the Forward CLTU service in terms of the state transitions of the Forward CLTU service provider;
- e) ANNEX A is a formal specification of Forward CLTU service data types, using the Abstract Syntax Notation One (ASN.1).
- f) ANNEX B lists all terms used in this document and identifies where they are defined;
- g) ANNEX C illustrates the relationship between SLE Service Data Units (SLE-SDUs) and SLE Protocol Data Units (SLE-PDUs);
- h) ANNEX D lists all acronyms used within this document;

- i) ANNEX E provides a conformance matrix that defines what capabilities must be provided for an implementation to be considered compliant with this Recommendation;
- j) ANNEX F contains a list of informative references.

1.5.2 SLE SERVICES DOCUMENTATION TREE

This Recommendation is based on the architectural model for cross support defined in the *SLE Reference Model* (reference [1]). It expands upon the concept of an SLE Transfer Service as interactions between SLE Mission User Entities (MUEs) and an SLE Transfer Service provider for the purpose of providing the CLTU Transfer Service.

This Recommendation is part of a suite of documents specifying the SLE services. The SLE Services constitute one of the three types of Cross Support Services:

- a) Part 1: SLE Services;
- b) Part 2: Ground Communications Services;
- c) Part 3: Ground Domain Services.

The basic organization of the SLE services documentation is shown in Figure 1-1. The specific SLE documents that are directly associated with this Recommendation are listed below.

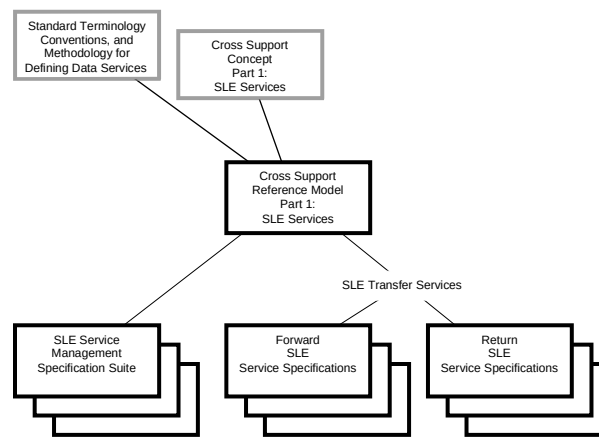


Figure 1-1: Space Link Extension (SLE) Services Documentation

- a) *Standard Terminology, Conventions and Methodology (TCM) for Defining Data Services*; a Report identifying the existing international standards for defining SLE data services, reference [F3].
- b) *Cross Support Concept – Part 1: Space Link Extension Services*; a Report introducing the concepts of cross support and SLE services, reference [F5];
- c) *Cross Support Reference Model — Part 1: Space Link Extension Services*; a Recommendation that defines the reference model which provides a common

terminology and framework for coordinating the development of CCSDS cross support recommendations, reference [1]

- d) *SLE Forward Transfer Service Specifications*; this Forward CLTU Transfer Service Recommendation is one of a set of Recommendations that will provide specification of all forward link SLE Transfer Services;
- e) *SLE Return Transfer Service Specifications*; a set of Recommendations that will provide specification of all return link SLE Services;
- f) *SLE Service Management Specification Suite*; a set of Recommendation that establish the basis for SLE service management. The primary Recommendation in this suite, the Service Management Specification, is listed as reference [F4].

1.6 DEFINITIONS, NOMENCLATURE, AND CONVENTIONS

1.6.1 DEFINITIONS

1.6.1.1 Definitions from OSI Basic Reference Model

This Recommendation makes use of a number of terms defined in reference [7]. The use of those terms in this Recommendation shall be understood in a generic sense; i.e., in the sense that those terms are generally applicable to technologies that provide for the exchange of information between real systems. Those terms are:

- a) abstract syntax
- b) application layer
- c) application process
- d) concatenation
- e) flow control
- f) Open System Interconnection (OSI)
- g) real system
- h) service access point (SAP)

1.6.1.2 Definitions from Abstract Syntax Notation One

This Recommendation makes use of the following terms defined in reference [6]:

- a) Abstract Syntax Notation One (ASN.1)
- b) object identifier

- c) (data) type
- d) (data) value

NOTE – In ANNEX A of this Recommendation, ASN.1 is used for specifying the abstract syntax of Forward CLTU service operations. The use ASN.1 as a descriptive language is intended to support the specification of the abstract Forward CLTU service; it is not intended to constrain implementations. In particular, there is no requirement for implementations to employ ASN.1 encoding rules. ASN.1 is simply a convenient tool for formally describing the abstract syntax of Forward CLTU service operations.

1.6.1.3 Definitions from Abstract Service Definition Conventions

This Recommendation makes use of two terms defined in reference [8]. The use of those terms in this Recommendation shall be understood in a generic sense; i.e., in the sense that those terms are generally applicable to technologies that provide for the exchange of information between real systems. Those terms are:

- a) initiator;
- b) responder.

1.6.1.4 Definitions from Directory Service - Models

This Recommendation makes use of the following terms defined in reference [9]:

- a) distinguished name (DN);
- b) relative distinguished name (RDN).

1.6.1.5 Definitions from Telecommand Channel Service

This Recommendation makes use of the following terms defined in reference [2]:

- a) Acquisition Sequence
- b) Command Link Transmission Unit (CLTU)
- c) Carrier Modulation Mode (CMM)
- d) Idle Sequence
- e) Physical Layer Operations Procedure (PLOP)

1.6.1.6 Definitions from Telecommand Data Routing Service

This Recommendation makes use of the following terms defined in reference [3]:

- a) Command Link Control Word (CLCW)

- b) Expedited (BD) Service
- c) Sequence Controlled (AD) Service

1.6.1.7 Definitions from SLE Reference Model

This Recommendation makes use of the following terms defined in reference [1]:

- a) abstract binding
- b) abstract object
- c) abstract port
- d) abstract service
- e) CLTU channel
- f) Forward CLTU service
- g) invoker
- h) Mission Data Operation System (MDOS)
- i) Mission User Entity (MUE)
- j) offline delivery mode
- k) online delivery mode
- l) operation
- m) performer
- n) physical channel
- o) service provider (provider)
- p) service user (user)
- q) sldu (SL-DU)
- r) SLE Complex
- s) SLE Complex Management
- t) SLE data channel

- u) SLE functional group (SLE-FG)
- v) SLE protocol data unit (SLE-PDU)
- w) SLE service data unit (SLE-SDU)
- x) SLE service package
- y) SLE System
- z) SLE transfer service instance
- aa) SLE transfer service production
- bb) SLE transfer service provision
- cc) SLE Utilization Management
- dd) space link
- ee) space link data channel
- ff) space link data unit (SL-DU)
- gg) space link session

1.6.1.8 Additional Definitions

For the purposes of this Recommendation, the following definitions also apply.

1.6.1.8.1 Association

An association is a cooperative relationship between an SLE service-providing entity and an SLE service-using entity. An association is formed by the exchange of SLE protocol control information through use of an underlying communication service.

1.6.1.8.2 Communications Service

A communications service is a capability that enables an SLE service-providing application process and an SLE service-using application process to exchange information.

NOTE – If an SLE service user and an SLE service provider are implemented using different communications services, then interoperability between them is possible only by means of a suitable gateway. Adherence to this Recommendation ensures, at least in principle, that it is possible to construct such a gateway.

1.6.1.8.3 Confirmed Operation

A confirmed operation is an operation that requires the performer to return a report of its outcome to the invoker.

1.6.1.8.4 Invocation

The invocation of an operation is the making of a request by an object (the invoker) to another object (the performer) to carry out the operation.

1.6.1.8.5 Parameter

A parameter of an operation is data that may accompany the operation's invocation or return.

NOTE – The term parameter is also used to refer to a mission-dependent configuration item used in production or provision of the service.

1.6.1.8.6 Performance

The performance of an operation is the carrying out of the operation by an object (the performer).

1.6.1.8.7 Port Identifier

A port identifier identifies a logical channel in a communications service. A port identifier contains all the addressing information necessary to make a connection possible.

NOTE – Port identifiers are technology-specific. For example, a port identifier might be the combination of an Internet Protocol (IP) network address and a Transmission Control Protocol (TCP) port number or the combination of an OSI network address and an associated set of service access points (SAPs).

1.6.1.8.8 Return

The return of an operation is a report, from the performer to the invoker, of the outcome of the performance of the operation.

1.6.1.8.9 Service Instance Provision Period

The service instance provision period is the time during which a service instance, i.e., the capability to transfer one or more SLE data channels of a given type, is provided.

1.6.1.8.10 Unconfirmed Operation

An unconfirmed operation is an operation that does not require a report of its outcome to be returned to the invoker by the performer.

1.6.2 NOMENCLATURE

The following nomenclature applies throughout this Recommendation:

- a) the words ‘shall’ and ‘must’ imply a binding and verifiable specification;
- b) the word ‘should’ implies an optional, but desirable, specification;
- c) the word ‘may’ implies an optional specification;
- d) the words ‘is’, ‘are’, and ‘will’ imply statements of fact.

1.6.3 CONVENTIONS

1.6.3.1 Specification of Operations

1.6.3.1.1 General

Section 3 of this Recommendation specifies the operations that constitute the Forward CLTU service. The specification of each operation is divided into subsections as follows:

1.6.3.1.2 Purpose Subsection

The Purpose subsection briefly describes the purpose and functioning of the operation. Additionally, it indicates whether the operation may be invoked by the user, provider, or both; whether the operation is confirmed or unconfirmed; and whether there are any constraints on when the operation may be invoked.

1.6.3.1.3 Invocation, Return, and Parameters Subsection

The Invocation, Return, and Parameters subsection describes the parameters associated with each operation, including their semantics. A table accompanying the description of each operation lists all parameters associated with the operation and, for both the invocation and return, whether the parameter is always present, always absent, or conditionally present.

For parameters that are conditionally present, the parameter description specifies the conditions for the presence or absence of the parameter. The condition is generally based on the value of another parameter in the same invocation or return; for example, the parameter `diagnostic` is present in a return if and only if the value of the parameter `result` is “negative result”. For a conditional parameter in a return, the condition may be based on the value of a parameter in the corresponding invocation.

In the table, the following convention is used to indicate whether a parameter is always present, always absent, or conditionally present:

M	Always present (Mandatory)
C	Conditionally present

Blank Always absent

NOTE – Even though a parameter may be characterized as always present, its description may specify that its value is permitted to be “null” or “unused” or the like.

1.6.3.1.4 Effects Subsection

The Effects subsection describes the effects an operation has on the invoker, the performer, the association between them, or any combination thereof. The details of how those effects occur or the mechanisms used are outside the scope of this Recommendation.

1.6.3.2 Typographic Conventions

Typographic conventions used in this Recommendation are described in the following paragraphs.

1.6.3.2.1 Operation Names

Names of CLTU service operations always appear in uppercase and begin with the characters ‘CLTU-’ (e.g., CLTU-TRANSFER-DATA).

1.6.3.2.2 Parameter Names

Names of parameters of CLTU service operations generally appear in lowercase and are always typeset in a fixed-width font (e.g., `invoker-credentials`).

1.6.3.2.3 Value Names

For the reader’s convenience, values of parameters are presented in this Recommendation by names. These names are always shown in single quotation marks (e.g., ‘no such service instance’). The actual value associated with the name is constrained by the type of the parameter taking on this value. The parameter type is specified in ANNEX A of this Recommendation.

NOTES

- 1 The name of a value does not imply anything about type. For example, the value ‘no such service instance’ has the appearance of a character string but might be assigned to a parameter whose type is ‘integer’.
- 2 Contrary to traditional publication style, sentence punctuation, such as a comma or a period, follows the closing quotation mark of a value or type name rather than preceding it.

1.6.3.2.4 State Names

This Recommendation specifies the states of Forward CLTU service providers. States may be referred to by number (e.g., state 3) or by name. State names are always shown in single quotation marks (e.g., 'active').

1.6.3.2.5 Data Type Definitions

Data type definitions for the CLTU service are presented in ANNEX A in the form of a set of ASN.1 modules. Regardless of the conventions used elsewhere in this Recommendation, the text of the ASN.1 modules is typeset entirely in a fixed-width font.

1.6.3.3 Other Conventions

This Recommendation uses the conventions specified in reference [1].

1.7 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All documents are subject to revision, and users of this Recommendation are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS Recommendations.

NOTE – a list of informative references forms ANNEX F of this Recommendation.

- [1] *Cross Support Reference Model—Part 1: Space Link Extension Services*. Recommendation for Space Data System Standards, CCSDS 910.4-B-1. Blue Book. Issue 1. Pasadena, California, USA: CCSDS, May 1996.
- [2] *Telecommand Part 1 — Channel Service*. Recommendation for Space Data System Standards, CCSDS 201.0-B-2. Blue Book. Issue 2. Washington, D.C.: CCSDS, November 1995.
- [3] *Telecommand Part 2 — Data Routing Service*. Recommendation for Space Data Systems Standards, CCSDS 202.0-B-2. Blue Book. Issue 2. Washington, D.C.: CCSDS, November 1992.
- [4] *Telecommand Part 2.1 — Command Operation Procedures*. Recommendation for Space Data System Standards, CCSDS 202.1-B-1. Blue Book. Issue 1. Washington, D.C.: CCSDS, October 1991.
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2 DESCRIPTION OF THE FORWARD CLTU SERVICE

2.1 OVERVIEW

The Forward CLTU service enables the user of the service to send Command Link Transmission Units (CLTUs) to a spacecraft, via an established forward space link channel. A forward space link channel is a physical channel carrying an asynchronous stream of CLTUs (reference [1]).

The CLTU service user submits CLTUs, encapsulated in Space Link Extension (SLE) Service Data Units (SLE-SDUs), by means of the CLTU-TRANSFER-DATA operation. Production of the Forward CLTU service by the provider entails processing the CLTUs transferred by the user through the necessary transformations to modulate the radio frequency (rf) carrier channel providing uplink communications with the spacecraft.

The Forward CLTU service transmits the CLTUs in the order in which they are submitted by the service user. The provider may perform checks to determine if the CLTU complies with applicable constraints, e.g., that the length of the CLTU is within the maximum size set by service management. However, the provider does not interpret, interrogate or modify the contents of a CLTU. CLTUs are transmitted bit for bit as received from the service user.

The operations defined in section 3 of this Recommendation enable a single Forward CLTU service user to interact with a Forward CLTU service provider to:

- a) establish an association between the user and the provider;
- b) send annotated CLTUs to the provider;
- c) obtain notifications and reports regarding status, configuration, and performance of the service;
- d) temporarily stop and later re-start the sending of CLTUs;
- e) release an association.

The Sequence Controlled (AD) and Expedited (BD) Services, as defined in the Command Operation Procedures Recommendation (reference [4]) are accomplished by higher layer SLE services.

The Forward CLTU service is provided only in the online delivery mode, as defined in reference [1]. The offline delivery mode is the subject of further study.

The provision of Forward CLTU service for access to one physical channel by one service user constitutes one instance of service. Only a single service instance of the Forward CLTU service may exist per physical channel at a time.

2.2 SPACE LINK EXTENSION REFERENCE MODEL

The Forward CLTU service is specified within the framework defined by the SLE Reference Model (reference [1]). The SLE Reference Model is based, in part, on the Abstract Service Definition Conventions (ASDC) defined in reference [8]. The following paragraphs summarize selected concepts from the SLE reference model.

2.2.1 ABSTRACT OBJECT

An abstract object is a functional entity that interacts with other abstract objects. Objects are of different types, which determine their function and behavior. Objects are characterized by their interfaces, which are called abstract ports, and the operations which are made available through those interfaces.

2.2.2 ABSTRACT SERVICE

An abstract service is the capability provided by a set of operations that an abstract object exposes at one or more of its abstract ports.¹

2.2.3 ABSTRACT BINDING

When two abstract ports of the same type have an association established between them such that an abstract service can be provided, the two ports are said to be bound. The act of establishing such an association is called abstract binding.

The terms initiator and responder are used to describe the interaction between two objects with respect to abstract binding. One object (the initiator) invokes a bind operation which is accepted (or rejected) by another object (the responder).

2.2.4 SERVICE USER/PROVIDER

An object that offers a service to another by means of one or more of its ports is called a service provider (provider). The other object is called a service user (user). An object may be a provider of some services and a user of others.

The terms user and provider are used to distinguish the roles of two interacting objects. In this Recommendation, when two objects are involved in provision of a service, the object closer to the space link is considered to be the provider of the service, and the object further from the space link is considered to be the user.

2.2.5 OPERATION

An operation is a procedure or task that one object (the invoker) can request of another (the performer) through a port pair bound within the terms of an agreement (reference [1], 1.6.1.8).

The terms invoker and performer are used to describe the interaction between two objects as the operations that constitute the service occur. One object invokes an operation that is

¹ The concept of an abstract service is to be distinguished from the concept of an (N)-service as defined in the OSI Basic Reference Model (reference [7]). The definition of (N)-service is in terms of the capability provided by one layer in the OSI architecture to the layer above it. The definition of abstract service is in terms of the capability provided by one abstract object to another abstract object. In a cross support scenario, where one Agency is providing an SLE service to another Agency, the object that provides the service typically is associated with one Agency, and the object that uses the service typically is associated with the other Agency.

performed by the other. For most services, each object invokes some operations and performs others.

2.3SERVICE MANAGEMENT

Before Forward CLTU service may be provided, it must be scheduled and configured, which is to say it must be managed. Service management determines the schedule of Forward CLTU service instances to be provided, the resources required to enable those service instances, and the initial configuration of all service instances and their supporting resources. SLE service management is the subject of separate CCSDS Recommendations; contact the CCSDS Secretariat for information regarding the status of those Recommendations.

Configuration parameters that are associated with the production of Forward CLTU service, such as bit rate and modulation index, potentially can impact SLE Complex resources. Consequently, these parameters are initialized by service management and can be modified only by service management. Configuration parameters that are associated with the provision of the service, such as the frequency at which status reports shall be generated, are initialized by service management but may be modified by the user through operations specified in this Recommendation.

Information required for access control, and for authentication of the service user by the provider and of the provider by the user, is provided via service management.

2.4ARCHITECTURE MODEL – FUNCTIONAL VIEW

2.4.1 FORWARD TC SPACE LINK PROCESSING SLE FUNCTIONAL GROUP

The Forward TC Space Link Processing SLE Functional Group (SLE-FG) shown in Figure 2-1 produces the Forward CLTU service.

As described in reference [1], the Forward TC Space Link Processing SLE-FG consumes one CLTU data channel consisting of a stream of CLTU SLE-SDUs and injects the CLTUs into the asynchronous physical channel.

NOTE – Per physical forward channel, only a single instance of the Forward CLTU service is supported at any point in time.

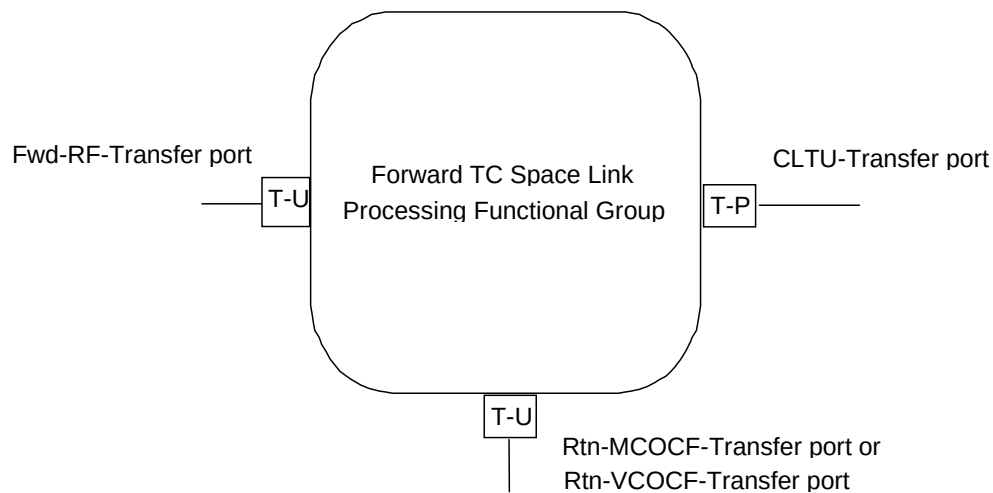


Figure 2-1: Forward TC Space Link Processing SLE-FG

The Forward TC Space Link Processing SLE-FG performs the following functions with respect to the Forward CLTU service:

- a) The Forward TC Space Link Processing SLE-FG consumes one CLTU data channel and extracts CLTUs encapsulated in SLE Service Data Units (SLE-SDUs).
- b) The Forward TC Space Link Processing SLE-FG also consumes one Operational Control Field (OCF) data channel and extracts the Command Link Control Words (CLCWs); based on the values in the CLCWs, the Forward CLTU service determines whether the physical channel is available.

NOTE – CLCWs may be ignored, as an option set by Service Management. See 3.1.10 and Table 3-11.

- c) The Forward TC Space Link Processing SLE-FG performs the following:
 - 1. generates Acquisition Sequence and Idle Sequence on the physical channel in accordance with the PLOP in effect;
 - 2. utilizes the underlying antenna steering capabilities provided by the ground element;
 - 3. modulates the CLTUs as a stream of bits on the ground-to-space RF channel; and
 - 4. radiates the signal to the spacecraft.

In addition to encapsulating the CLTUs, the SLE-SDUs contain control and annotation data that specify radiation time and other parameters to aid in processing the data. The SLE-SDUs

containing CLTUs are described in section 3.6, which defines the CLTU-TRANSFER-DATA operation.

2.4.2 FORWARD CLTU SERVICE PRODUCTION AND PROVISION

Forward CLTU **production** is concerned with radiating CLTUs extracted from a stream of SLE-SDUs according to the control and annotation of the CLTU in the SLE-SDU and according to the configuration set up by service management. Forward CLTU service **provision** is concerned with receiving a stream of SLE-SDUs from a Forward CLTU service user. Service provision addresses such matters as when service is provided (e.g., service start and stop times), and how service is provided (e.g., which events are notified to the user).

The SLE-SDUs consumed by the Forward CLTU service are sent by the service user by means of the Forward CLTU service operations defined in section 3. These operations also provide additional functionality to facilitate the provision of service, i.e. enabling the exchange of SLE-SDUs across a remote interface. The service operations are realized as SLE protocol data units (SLE-PDUs) which are exchanged between the Forward CLTU service provider and the Forward CLTU service user by means of an underlying communications service. The general relationship between SL-DUs, SLE-SDUs, and SLE-PDUs is illustrated in Figure 2-2. Refer to ANNEX C for a fuller discussion of this topic.

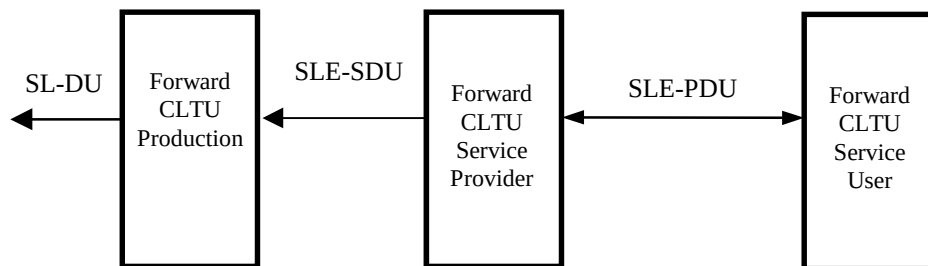


Figure 2-2 Forward CLTU Service Production and Provision

Production of the Forward CLTU service by the provider occurs during the space link session; in general, production will largely overlap with service provisioning.

2.5 ARCHITECTURE MODEL – CROSS-SUPPORT VIEW

The management and control of the production and provisioning of the SLE transfer services is specified in reference [1]. Figure 2-3 shows an example of Forward CLTU SLE Complex and the binding of its Transfer and Management Ports. This example shows an SLE Complex with one Forward TC Space Link Processing FG instance; it is providing one instance of Forward CLTU service to Mission User Entities (MUEs) in a Mission Data Operations System (MDOS). As this figure shows, only a single Mission User Entity (MUE) can use the Forward CLTU service provided by a single Forward TC Space Link Processing SLE-FG.

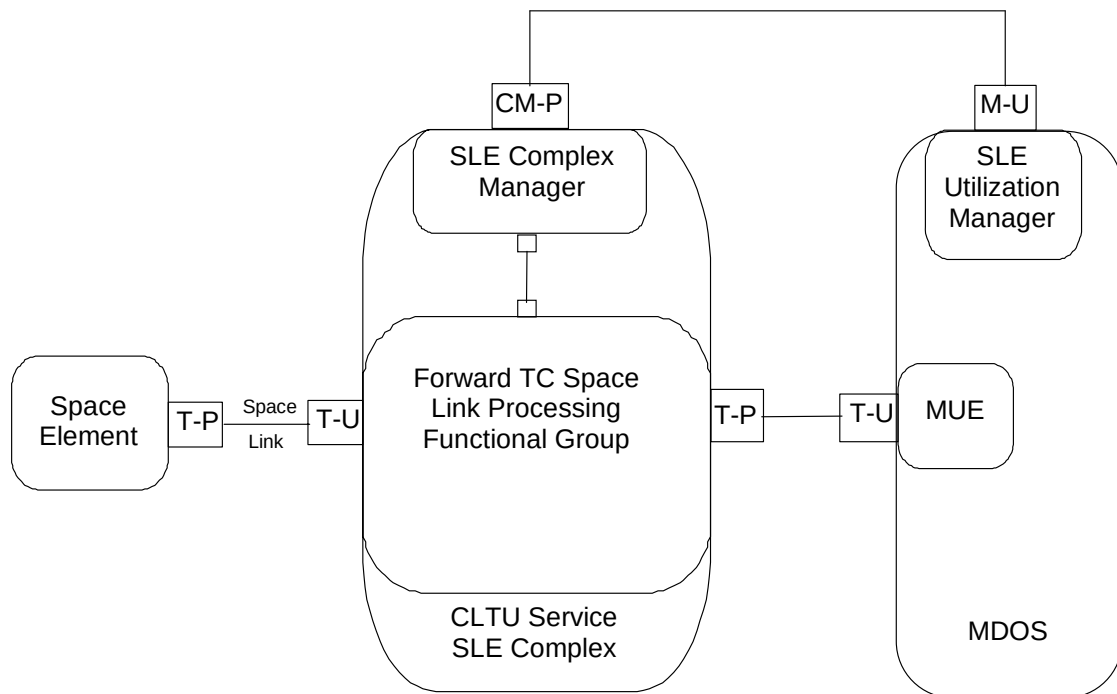


Figure 2-3: An Example of Binding of the Forward CLTU Service SLE Complex

NOTE – The Forward CLTU service occurs between the MDOS T-U port and the SLE Complex T-P port.

2.6FUNCTIONAL DESCRIPTION

2.6.1GENERAL

This subsection describes the Forward CLTU service with respect to scheduling, configuration, provider states and protocol considerations.

2.6.2CONFIGURATION AND SCHEDULING

The management of the SLE Complex that is providing the Forward CLTU service negotiates with SLE Utilization Management of the MDOS to establish mutually agreed upon SLE service packages. Among other things, SLE service packages specify the services to be provided, the schedules of service instances, and the resources needed to enable those services.

Service packages also specify the initial values of the mission-dependent parameters required for service production and provision. Forward CLTU service production parameters include such things as bit rate, modulation index, and subcarrier frequency. Provision parameters include such things as scheduled start and stop times of the Forward CLTU service instance.

Service production is guaranteed to occur only as needed to support service packages that have been scheduled and mutually agreed upon by SLE Complex Management and SLE

Utilization Management. Service provision occurs only within the bounds of the agreed upon schedule of service instances and only during those periods when there is an association between the service provider and the service user.

2.6.3BUFFERING

The Forward CLTU service buffers the CLTUs, for the primary purpose of maintaining radiation of a steady stream of CLTUs despite changes in uplink bit rates and variable latency over the ground communications channel. All transfers of CLTUs from the service user to the service provider must occur within the scheduled service instance provision period.

2.6.4PROTOCOL DESCRIPTION

2.6.4.1CLTU Operations

The operations that constitute the CLTU service are listed in Table 2-1. Section 3 of this Recommendation contains the detailed specification of these operations.

Table 2-1: CLTU Operations

Service Operation	Invoked By	Con-firmed?	Purpose
CLTU-BIND	Initiator (User)	Yes	To establish an association with the provider
CLTU-UNBIND	Initiator (User)	Yes	To release an association previously established by a CLTU-BIND operation
CLTU-START	User	Yes	To request that the SLE service provider prepare to accept CLTU-TRANSFER-DATA operations
CLTU-STOP	User	Yes	To request that the SLE service provider stop service provision and production.
CLTU-TRANSFER-DATA	User	Yes	To transfer a CLTU to the service provider
CLTU-ASYNC-NOTIFY	Provider	No	To notify the user of an event affecting production or provision of the CLTU service
CLTU-SCHEDULE-STATUS-REPORT	User	Yes	To request that the provider send a status report immediately or periodically, or stop reporting
CLTU-STATUS-REPORT	Provider	No	To send a status report to the user
CLTU-GET-PARAMETER	User	Yes	To ascertain the value of an SLE service parameter (see Table 3-1 and Table 3-11).
CLTU-THROW-EVENT	User	Yes	To forward an event that requires Complex Management to take the actions defined for this event
CLTU-PEER-ABORT	User or Provider	No	To notify the peer system that the local system detected an error that requires the association to be terminated

2.6.4.2 States of the Service Provider

Once created, the Forward CLTU service instance is in one of three states, as follows:

- a) State 1 ('unbound'): In state 1, all resources required to enable the provision of the Forward CLTU service have been allocated, and all objects required to provide the service have been instantiated. However, no association yet exists between the user and the provider (i.e., the Forward CLTU transfer service provider port is not bound).
- b) State 2 ('ready'): In state 2, an association has been established between the user and the provider, and they may interact by means of the operations described in section 3 of this Recommendation. However, sending of CLTUs from the user to the provider (by means of the CLTU-TRANSFER-DATA operation) is not permitted. The user may enable the delivery of CLTUs by means of the appropriate service operation (CLTU-START), which, in turn, will cause the provider to transition to state 3 ('active').
- c) State 3 ('active'): State 3 resembles state 2 ('ready') except that now the user can send CLTUs and the provider is enabled to radiate CLTUs to the spacecraft. The

service continues in this state until the user invokes the CLTU-STOP operation to cause the provider to suspend transmission of CLTUs and transition back to state 2.

A simplified state transition diagram for the CLTU service provider is shown in Figure 2-4. A detailed state transition matrix is provided in section 4.

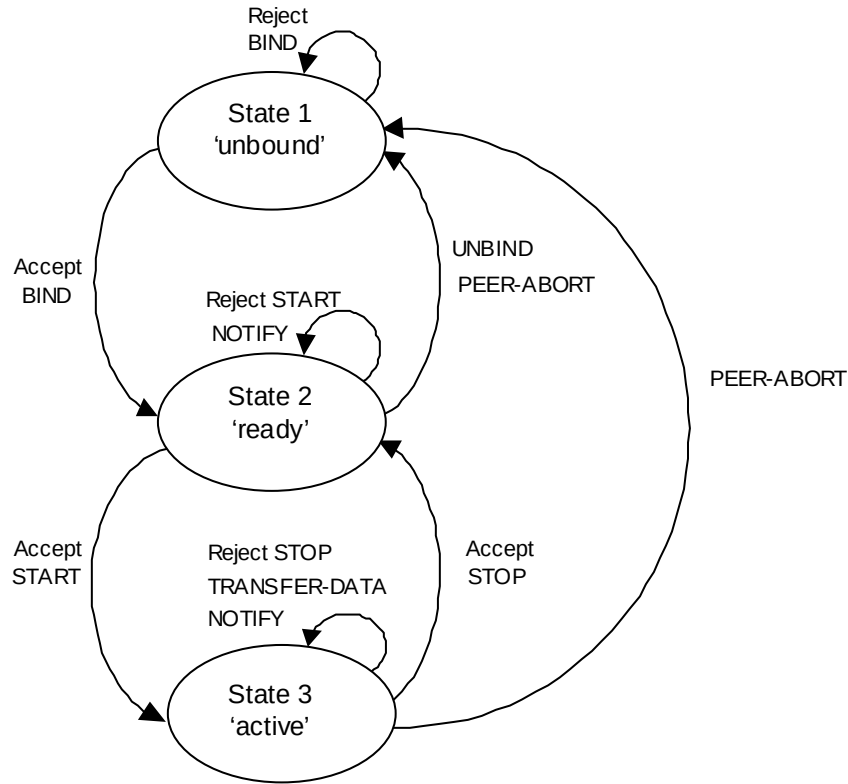


Figure 2-4: Forward CLTU Service Provider State Transition Diagram

2.6.4.3 Terminating an Association

An association is released normally when a CLTU-UNBIND is issued by the user (the initiator of the association) and accepted by the provider. An association may be aborted by either the user or the provider by means of the CLTU-PEER-ABORT operation. An association may also be aborted due to a failure in the underlying communications system. Such failures are signaled to the local application by the 'protocol abort' event (see 4.1.5).

2.6.4.4 Effects of Association Termination

The production of CLTUs stops immediately following the termination of an association, except for a CLTU in the process of being radiated. Any buffered CLTUs are discarded.

When an association is terminated no further operations can be exchanged between the user and the provider. The systems may re-establish an association via a new CLTU-BIND operation. Status information is not preserved after an association terminates, and is not available to the new association with the following exceptions:

- a) all accounting information, such as the number of CLTUs processed, is persistent for the duration of the service instance provision period.
- b) parameters that serve to relate notifications on an activity to operations that triggered this activity will not be altered when the association is released or aborted.

2.6.4.5 Technology Aspects

This Recommendation defines the Forward CLTU service. Provision of the Forward CLTU service in a real system also requires a specification of how the service is mapped to a communications service, such that the invocations and returns of the Forward CLTU service operations can be exchanged between the user and the provider. In order not to restrict the applicability of this Recommendation to a specific communications technology, as few assumptions as possible have been made about the characteristics of the underlying communications service (see 1.3.1).

The service interface between the user and the provider is specified in this Recommendation in terms of the operations that the service provides. The implementation of the service in a real system requires a mapping of the service operations to protocol data units that can be exchanged by means of the communications service. There is a conceptual distinction between Forward CLTU service operations and SLE-PDUs; however, the mapping of service operations and returns to SLE-PDUs illustrated in Figure 2-5 is one-to-one.

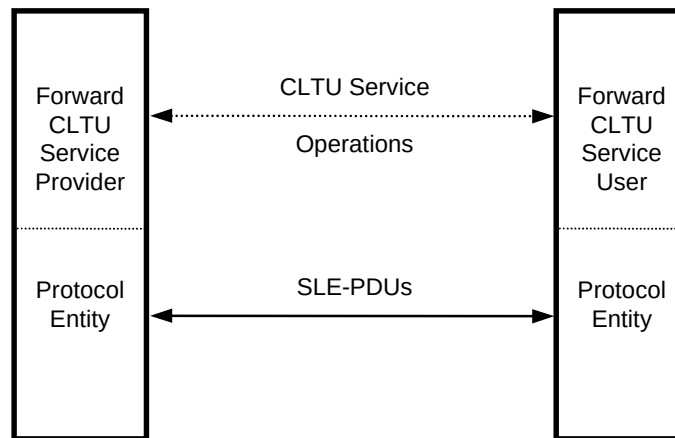


Figure 2-5: Mapping of Forward CLTU Service Operations to SLE-PDUs

The situation is complicated further when the underlying communications service is considered. In the case where the peer applications have been built on top of different communication technologies, it is possible to achieve interoperability by means of a gateway. Figure 2-6 shows a user and a provider that use different communications technologies and interoperate through a gateway. The Forward CLTU SLE-PDUs are translated on one side into forms appropriate for a particular communications technology. These forms are then

translated by the gateway into forms appropriate for another communications technology. Finally, those forms are translated back into Forward CLTU SLE-PDUs.

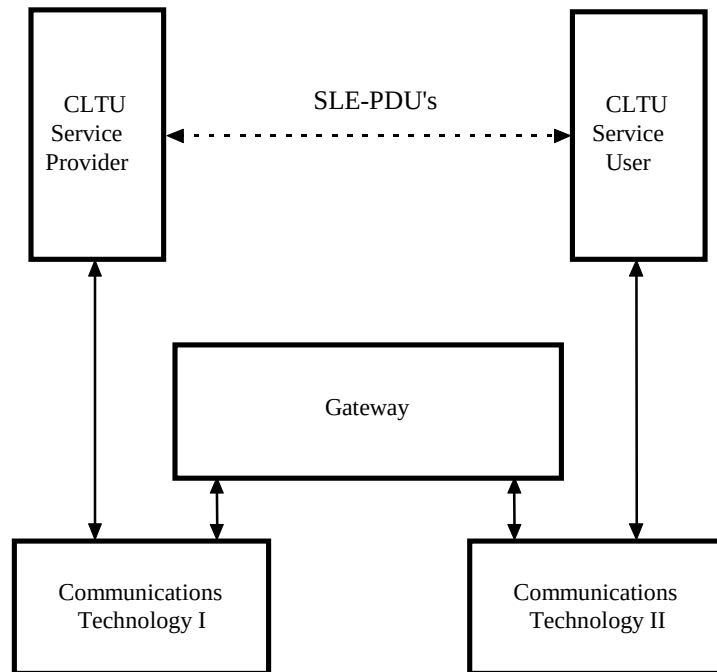


Figure 2-6: Forward CLTU Service Provision Via a Gateway

NOTE – In order to accomplish interoperability, even in the case of implementations using the same underlying communications technology, there is still the need to agree on how to map the SLE-PDUs into forms appropriate for that communications technology.

Because the operations of the Forward CLTU service are relatively simple, once an association is in place between the service user and the service provider, the technology-specific elements involved in the exchange of SLE-PDUs are relatively minor. However, the way an association is established (i.e., the binding) tends to be specific to the communications technology in use. Nonetheless, the CLTU-BIND and CLTU-UNBIND operations as specified in this document are intended to be ‘technology neutral’. This neutrality necessitates some flexibility in the specification of the parameters of the bind operation. The semantics and even the type of some of the bind parameters may vary depending on the communications technology that is used; any such differences must be reconciled by the gateway. The following paragraphs discuss these technology-dependent parameters.

NOTE – Conceptually, it is useful to view the gateway as being part of either the SLE Complex or the MDOS. In terms of management, the parameters relevant for the binding are communicated in the form of the communications technology employed by the system that does not include the gateway. It is the responsibility of the gateway-hosting system to convert the parameters submitted by management to the ‘local’ form.

The `responder-port-identifier` parameter identifies the logical name of the location of the peer SLE application to which the initiator is requesting to bind.

After conversion to the technology specific form, the responder port identifier serves a dual purpose:

- a) It identifies a ‘physical’ address that must be known so that the communications provider can route the SLE-PDUs as required. If a gateway is involved, the conversion of the logical name at the initiating side must deliver the ‘physical’ address of the gateway. As part of the conversion by the gateway the gateway’s ‘physical’ address will be replaced with the address of the responding SLE system.

NOTE – The term ‘physical’ address is used here in a generic sense and not necessarily limited to the identification of a physical port. Depending on the technology, it may also encompass elements like a port number (TCP), selectors (OSI transport and higher layers), etc.

- b) The responder port identifier also identifies the responder application’s location. A gateway may have to adapt this to the technology-specific presentation expected at the receiving end, but logically this part of the port identifier will not be modified by a gateway.

Binding is accomplished only when the association between the peer applications is established. A communications gateway implementation must ensure this. In particular, it must only return a report of the outcome of the CLTU-BIND operation to the initiator when it has received such a return from the responder. It is not sufficient to establish an association between the initiator and the gateway.

A gateway must also ensure that the state of the application protocol is kept consistent on the user and the provider sides. In particular, in case the association with the Forward CLTU application on one side of the gateway is aborted (e.g., because of a communications service fault), the gateway must ensure that the association with the Forward CLTU application on the other side of the gateway is also aborted.

2.7 OPERATIONAL SCENARIO

The following illustrates a typical sequence of operations between the user and the provider of the Forward CLTU service. A complete definition of the operations is found in section 3; the formal specification of provider behavior is presented in section 4.

- a) The user performs the CLTU-BIND operation to establish an association. The provider, when configured to monitor the No RF Available and/or No Bit Lock flags returned from the spacecraft in the Command Link Control Word (CLCW), performs

the necessary operations to receive the Operational Control Field (OCF) as provided by the Return Frame Processing SLE-FG.

- b) The provider generates the Acquisition and Idle Sequences on the physical channel in accordance with the physical link operational procedure (PLOP) in effect, when one of two conditions is met: either RF lock on the uplink signal has been achieved at the spacecraft (i.e., the No RF Available flag received in the CLCW is false); or, the `rf-available-required` parameter is not set.
- c) The user sends CLTU-START, and the provider goes to state 3, 'active'.
- d) The user sends a CLTU-TRANSFER-DATA operation to the provider. The provider verifies the CLTU, and if acceptable, buffers it until the specified `radiation-start-time` is reached.
- e) Additional CLTUs may be sent by the user and buffered.
- f) At the time specified for start of radiation, the first CLTU is injected into the physical channel and modulated onto the RF carrier. The signal is radiated to the spacecraft. If no start time was specified in the CLTU-TRANSFER-DATA operation, the CLTU is radiated as soon as received, or whenever the provider is ready to radiate, if that is later.
- g) Successive CLTUs are processed in similar fashion.
- h) The user transfers the last CLTU to the provider.
- i) The provider completes processing the buffered CLTUs. When the provider's CLTU buffer is empty, it sends CLTU-ASYNC-NOTIFY to inform the user.
- j) The user sends CLTU-STOP and the provider transitions to state 2, 'ready'. If PLOP-2 is in effect, the provider continues to radiate idle sequence. If PLOP-1 is in effect, the provider has already dropped command modulation and continues to radiate unmodulated carrier only (CMM-1).
- k) The user performs CLTU-UNBIND to release the association.
- l) At the end of the scheduled service period, the provider transitions to CMM-1 if PLOP-2, then typically turns off the RF link.

3SERVICE OPERATIONS AND THEIR PARAMETERS

3.1GENERAL CONSIDERATIONS

NOTES

1. This subsection specifies the processing of valid (i.e., recognized) SLE-PDUs. It defines common aspects of the operations of the Forward CLTU service.
2. Handling of invalid SLE-PDUs is specified in 4.1.

3.1.1RESULT OF OPERATIONS

3.1.1.1 All confirmed operations shall report on the outcome of the operation in a return.

3.1.1.2 All returns shall include a `result` parameter that indicates whether the outcome of the operation was successful ('positive result') or unsuccessful ('negative result').

3.1.1.3 In the event of a 'negative result', the return shall also include a `diagnostic` parameter that is descriptive of the reason for the 'negative result'.

NOTE – Possible values of the `diagnostic` parameter are listed in the description of each operation.

3.1.1.4 A `diagnostic` of 'other' shall be returned only if no other `diagnostic` in the list adequately describes the reason for the 'negative result'.

3.1.2PARAMETER TYPES

The types of all parameters shall conform to the abstract syntax specified in ANNEX A.

NOTE – Some parameter types in ANNEX A are chosen such that possible future extensions of the range of allowed values of a parameter will not cause a type mismatch. For example, parameters that logically are of the 'enumerated' type are specified as being of the 'named integer' type.

3.1.3PARAMETER CHECKING

3.1.3.1 Validity checks shall be performed on the values of parameters associated with operations.

NOTES

- 1 A parameter is invalid if its value is outside the range or not in the set of values currently permitted by service management for the given parameter. A conformant implementation is however expected to be capable to support the full range or set as specified in ANNEX A of this Recommendation.
- 2 A parameter is also invalid if its value is in conflict with the value of another parameter in the same invocation (e.g., if in CLTU-TRANSFER-DATA the time specified in the `earliest-radiation-time` parameter is later than the time specified in the `latest-radiation-time` parameter).

3.1.3.2 If a parameter is not valid, the operation shall not be performed, and, for confirmed operations other than CLTU-UNBIND, a report of ‘negative result’ shall be returned to the invoker.

3.1.3.3 Except as noted in 3.2.2.11, checks for invalid parameters or for other conditions that can cause a report of ‘negative result’ should be performed in the order in which diagnostics are listed in the descriptions of the operations.

3.1.3.4 In event of ‘negative result’, the **diagnostic** parameter shall be set to the value defined for the first problem found.

3.1.3.5 In the case that an implementation does not adhere to the sequence of checks as specified by the sequence of diagnostic values, such implementation shall specify the sequence in which checks are actually performed.

3.1.4 ACCESS CONTROL

3.1.4.1 The Forward CLTU service shall implement access control based on the identity of the initiator and responder. Access control is performed at two levels:

- a) The initiator must be registered at the responder and the responder must be registered at the initiator;
- b) The initiator and responder must be authorized for the given service instance.

3.1.4.2 The initiator shall maintain a registry of authorized responders and the responder shall maintain a registry of authorized initiators. These registries are controlled by SLE Complex Management and SLE Utilization Management as applicable.

3.1.4.3 Service management shall specify the authorized initiator and responder for each service instance.

3.1.4.4 The initiator and responder shall indicate their identity by setting the parameters **initiator-identifier** and **responder-identifier** in the CLTU-BIND operation to the values assigned by service management.

3.1.5 AUTHENTICATION

NOTE – Requirements for security depend on the application and on the SLE system environment (e.g., whether closed or public networks are used or if access is only from physically restricted areas). In many environments, security may be provided by the communications service, transparently to the SLE application. This Recommendation does not preclude the use of security features that are provided by the communications service or the local environment, nor does it assume the availability of such features.

3.1.5.1 The Forward CLTU service shall provide the following options with respect to level of authentication:

- a) “all”: authentication of all Forward CLTU operations that carry credentials parameters;

- b) “bind”: authentication of the CLTU-BIND operation, only;
- c) “none”: no authentication provided.

3.1.5.2 SLE Complex Management and SLE Utilization Management shall agree on the level of authentication to be required for an association between a user and the Forward CLTU service provider and configure both entities accordingly.

3.1.5.3 SLE Complex Management and SLE Utilization Management shall agree on the algorithms used to generate and check credentials parameters and make this algorithm known to the service user and provider, together with associated parameters such as passwords or keys as necessary for the adopted algorithm.

NOTES

1. The specification of the algorithms themselves is outside the scope of this Recommendation.
2. The `initiator-identifier` and `responder-identifier` parameters of the CLTU-BIND identify the user and provider respectively, and therefore the applicable authentication level and the algorithm(s) necessary to generate and check credentials.

3.1.5.4 For operations for which authentication of credentials is required by terms of the agreement between SLE Complex Management and SLE Utilization Management:

- a) invocations shall include an `invoker-credentials` parameter to permit the performer to authenticate the invocation; and
- b) returns shall include a `performer-credentials` parameter to permit the invoker to authenticate the return.

3.1.5.5 For operations for which authentication of credentials is not required, the `invoker-credentials` or `performer-credentials` parameter may be set to the value ‘unused’ to signify that the invocation or return does not carry credentials.

3.1.6 THREADED APPLICATIONS

3.1.6.1 To support applications that may be implemented using multiple threads of execution, the `invoke-ID` parameter shall be specified for all confirmed operations except CLTU-BIND and CLTU-UNBIND. The invoker must always block on CLTU-BIND, and no additional operations may be invoked after CLTU-UNBIND.

NOTE – The `invoke-ID` parameter allows the invoker to correlate a particular return to the invocation that prompted it.

3.1.6.2 After invoking an operation, the invoker may invoke another operation without waiting for the return from the first invocation.

3.1.6.3 The `invoke-ID` parameter shall contain an invoker-supplied arbitrary integer value that shall be returned, unchanged, by the performer.

3.1.6.4 An error condition shall exist if an invocation includes an `invoke-ID` whose value is the same as that of another invocation that is awaiting confirmation.

3.1.6.5 To ensure that the CLTU service behaves in a predictable manner, the effects of operations shall be as though the operations were performed in the order that their invocations were received by the performer.

3.1.6.6 The invoker may opt not to exploit the non-blocking capability and always wait for the return before invoking the next operation.

NOTE – An invoker wishing to operate in blocking mode, i.e., to invoke a new operation only when the return from the previous operation was already received, can provoke a blocking performer behavior by using a constant value for the `invoke-ID` parameter. As long as a return is still outstanding, the performer will reject any further invocation.

3.1.6.7 Compliance with this Recommendation does not require the performer to process invocations concurrently; however, the performer must still accept invocations in non-blocking mode and buffer and serialize them by local means not visible externally.

3.1.7 MANDATORY PARAMETERS

All providers shall be able to accept and respond to parameters specified as mandatory in ANNEX E.

NOTE – All parameters of operations are mandatory except where stated differently in ANNEX E. The actual presence of a parameter in a particular invocation or return may be conditional based on the value of another parameter in the same invocation or return. For some parameters, a value of 'null' is allowed, as discussed in the description of the relevant operations.

3.1.8 TIME

Universal Time Coordinated (UTC) shall be used for all parameters containing a time value. Time values shall be expressed to the tenth (0.1) of a second; values may be expressed to greater precision, as an implementation option.

3.1.9 DELIVERY MODES

3.1.9.1 Forward Online Delivery

Forward online delivery service provisioning occurs at the same time as service production, i.e., during a space link session. CLTUs supplied by the service user are buffered by the service provider until they are processed. The buffer used by the service provider exists only during service production. The timing of CLTU processing is determined by the order of CLTUs in the buffer and any annotation data provided with the CLTUs.

The forward online delivery mode is defined in this Recommendation.

3.1.9.2 Forward Offline Delivery

Service provisioning and service production do not overlap. CLTUs supplied by the service user during service provision are buffered by the provider in persistent storage until service production.

The forward offline delivery mode is not defined in this version of this Recommendation.

3.1.10 SETTING OF PARAMETERS

A Forward CLTU provider shall permit setting of the parameters as specified in Table 3-1. The range or set of values a parameter may assume is constrained by specification of its data type (see ANNEX A). Service management may further constrain the allowed values for a given service instance. Further details on these parameters are presented in Table 3-11.

NOTE – Table 3-1 contains only parameters for which the user can ascertain the current value by means of the CLTU-GET-PARAMETER operation. A complete list of parameters that may affect service production is to be found in [F4].

Table 3-1: Setting of Forward CLTU Service Parameters

Parameter Control	Service Management	CLTU-START Operation	CLTU-SCHEDULE-STATUS-REPORT Operation	CLTU-THROW-EVENT Operation
bit-lock-required	X			
delivery-mode	X			
expected-cltu-identification		X		
expected-event-invocation-identification				X
maximum-cltu-length	X			
modulation-frequency	X			
modulation-index	X			
plop-in-effect	X			
reporting-cycle			X	
return-timeout-period	X			
rf-available-required	X			
subcarrier-to-bit-rate-ratio	X			

3.1.11 PROVIDER BUFFERING REQUIREMENTS

3.1.11.1 The service package shall specify the amount of buffering the provider must maintain.

3.1.11.2 The amount of buffer space shall be specified in terms of the number of octets that can be stored.

3.1.11.3 The service provider shall buffer only complete CLTUs.

3.1.12 ACCOUNTING SUMMARY

Statistical information to be collected over a period of time shall always refer to the service instance provision period.

3.2CLTU-BIND

3.2.1PURPOSE

3.2.1.1 The initiator shall invoke the CLTU-BIND operation to establish an association between the initiator and responder as defined in 1.6.1.8.1.

3.2.1.2 The responder shall confirm the CLTU-BIND operation.

3.2.1.3 Except as provided in 3.2.1.4, the initiator shall not invoke any further CLTU operations on this service instance until the bind is confirmed.

3.2.1.4 If the return from the invocation of CLTU-BIND is not received after a sufficiently long time (to be determined by service management), the initiator may attempt to recover by invoking the CLTU-PEER-ABORT operation (see 3.12) followed by another CLTU-BIND.

3.2.1.5 The CLTU-BIND operation is valid only in state 1 ('unbound') and shall be invoked only by the user.

3.2.2INVOCATION, RETURN, AND PARAMETERS

3.2.2.1General

The parameters of the CLTU-BIND operation shall be present in the invocation and return as specified in Table 3-2.

Table 3-2: CLTU-BIND Parameters

Parameter	Invocation	Return
invoker-credentials	M	
performer-credentials		M
initiator-identifier	M	
responder-identifier		M
responder-port-identifier	M	
service-type	M	
version-number	M	C
service-instance-identifier	M	
result		M
diagnostic		C

3.2.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-BIND invocation (see 3.1.5).

3.2.2.3performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of CLTU-BIND (see 3.1.5).

3.2.2.4initiator-identifier

The **initiator-identifier** parameter shall identify the authority on whose behalf the SLE application is initiating an association.

NOTES

- 1 The **initiator-identifier** parameter permits the responding SLE application to determine if this user is registered at it and which authentication scheme is to be applied.
- 2 In case authentication based on credentials is used, this information may be redundant since the initiator-identifier value is normally one constituent of the **initiator-credentials** parameter. However, its encoding may differ and it may be convenient to have this parameter also available in “clear text” form.

3.2.2.5responder-identifier

The **responder-identifier** parameter shall identify the authority on whose behalf the responding SLE application is acting.

NOTES

- 1 The **responder-identifier** parameter permits the initiator to determine if the responder from which the CLTU-BIND return originates is registered at this initiator.
- 2 The initiator uses this parameter, if applicable after having successfully authenticated the CLTU-BIND return to determine if this return originates from the intended responder.

3.2.2.6responder-port-identifier

The **responder-port-identifier** parameter shall specify the logical name of the responding SLE application’s location the initiator wishes to connect to. When such logical name is assigned, the information necessary to convert this logical name to an address permitting to establish a connection with the named location must be provided as well.

NOTE – The method used to convert the logical names to the communications technology specific port addresses is outside the scope of this Recommendation (see 2.6.4.5).

3.2.2.7 **service-type**

The **service-type** parameter shall specify the type of service that will be present if the bind operation succeeds.²

3.2.2.8 **version-number**

3.2.2.8.1 The **version-number** parameter shall identify the version number of the CLTU service specification that is to govern this association if the CLTU-BIND succeeds.

3.2.2.8.2 **version-number** is conditionally present in the return based on the **result** parameter:

- a) if the value of **result** is 'positive result', **version-number** shall be present in the return;
- b) if the value of **result** is 'negative result', **version-number** shall not be present.

3.2.2.8.3 If the value of **result** is 'positive result', the responder shall either:

- a) accept the version proposed by the initiator by putting the same version number into the positive return; or
- b) if the responder implementation supports version negotiation, propose a lower (earlier) version number by putting the lower version number in the positive return.

3.2.2.8.4 If the responder implementation does not support version negotiation or does not support the requested version or a lower version, the responder shall reject the bind with the **diagnostic**, 'version not supported'.

3.2.2.8.5 If the initiator does not support the version proposed in the return, or the initiator implementation does not support negotiation, and the responder has proposed a lower version than the initiator requested, the initiator shall unbind the association.

3.2.2.8.6 The **version-number** value of the CLTU service defined by this issue of this Recommendation shall be '1'.

NOTE – The version negotiation process as outlined above is only feasible as long as a future version of the CLTU service retains the specification of the CLTU-BIND operation.

3.2.2.9 **service-instance-identifier**

The **service-instance-identifier** parameter shall uniquely identify this service instance within the scope of the service-providing SLE Complex.

² For the CLTU-BIND operation, the **service-type** parameter is redundant, because the only valid value of **service-type** is 'fwdCltu'. However, it is anticipated that future work by CCSDS will result in CLTU-BIND being superseded by a generic SLE-BIND operation that is invoked with any one of several SLE service types. The CLTU-BIND **service-type** parameter is provided in an attempt to facilitate such a change.

NOTE – It is recommended that **service-instance-identifier** take the form of a Distinguished Name (see reference [9]). Among other things, it identifies the SLE service agreement and the SLE service package applicable to this instance of service. More details on the way the service-instance-identifier is built can be found in [F4].

3.2.2.10 **result**

3.2.2.10.1 The **result** parameter shall specify the result of the CLTU-BIND invocation and shall contain one of the following values:

- a) ‘positive result’—the CLTU-BIND invocation is accepted by the responder and the association is established;
- b) ‘negative result’—the CLTU-BIND invocation is rejected by the responder for the reason specified in the **diagnostic** parameter.

3.2.2.11 **diagnostic**

3.2.2.11.1 If **result** is ‘negative result’, **diagnostic** shall be present and shall contain one of the following values:

- a) ‘access denied’—a user with the **initiator-identifier** value presented in the CLTU-BIND invocations is not registered at the performer;
- b) ‘service type not supported’— the **service-type** value in the CLTU-BIND invocation does not identify a service type supported by the responder;
- c) ‘version not supported’—the responder does not support the requested version and the responder implementation does not permit version negotiation or the responder does not support any version of the service lower than the one requested by the initiator;
- d) ‘no such service instance’—the requested service instance is not defined by any agreed upon service package known to the responder;
- e) ‘already bound’—the service instance is already bound via a different association;
- f) ‘service instance not accessible to this initiator’—the service instance identified by the value of **service-instance-identifier** parameter has not been created for use by the initiator specified in the **initiator-identifier** parameter;
- g) ‘inconsistent service type’—the **service-type** value in the CLTU-BIND invocation is not consistent with the service type of this service instance; i.e., it is not ‘fwdCltu’;
- h) ‘invalid time’—the CLTU-BIND operation was invoked outside the service instance provision period agreed to in the service package;
- i) ‘out of service’—the responder has been taken out of service for an indefinite period by management action;

- j) 'other'—the reason for the negative result will have to be found by other means.

NOTES

- 1 In some implementations, it may be inappropriate or impossible for the intended performer to provide a return in case of the conditions indicated by diagnostic d).
- 2 Implementations should consider that, under some conditions, CLTU-BIND might fail with no return, e.g., if `responder-port-identifier` has an incorrect value.

3.2.2.11.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.2.3EFFECTS

3.2.3.1 If `result` is 'positive result':

- a) an association between the user and the provider shall be established;
- b) the provider shall transition from state 1 ('unbound') to state 2 ('ready');
- c) upon receipt of the positive return, the user may proceed to invoke other CLTU service operations to initialize the service and enable data transfer (e.g., CLTU-SCHEDULE-STATUS-REPORT and CLTU-START).

3.2.3.2 If `result` is 'negative result':

- a) the association between the user and the provider shall not be established;
- b) the provider shall remain in state 1 ('unbound');
- c) upon receipt of the negative return:
 - 1) the initiator should examine the `diagnostic` parameter for the cause;
 - 2) the initiator may attempt to re-invoke the CLTU-BIND.

3.3CLTU-UNBIND

3.3.1PURPOSE

3.3.1.1 The initiator shall invoke the CLTU-UNBIND to release an association previously established by CLTU-BIND (see 3.2).

3.3.1.2 The responder shall confirm the CLTU-UNBIND operation.

3.3.1.3 Except as provided in 3.3.1.4, the initiator shall not invoke any further CLTU operations, including returns, on this service instance until the unbind is confirmed.

NOTE – The initiator may invoke the CLTU-UNBIND operation even if it did not yet receive all returns from previously invoked operations. The initiator should be aware that the responder may opt not to send any further returns as soon as it has received the CLTU-UNBIND invocation. It may then happen that the CLTU-UNBIND return is not received before one of the missing returns causes a ‘missing return’ exception (see 4.1.3) which in turn causes the association be aborted.

3.3.1.4 If the return from the CLTU-UNBIND invocation is not received after a sufficiently long time (to be determined by service management), the initiator shall invoke the CLTU-PEER-ABORT operation (see 3.12).

NOTE – The “sufficiently long time” as determined by service management can be ascertained by means of the CLTU-GET-PARAMETER operation and selecting ‘return-timeout-period’ as value of `cltu-parameter`.

3.3.1.5 The CLTU-UNBIND operation is valid only in state 2 ('ready') and shall be invoked only by the user.

3.3.2INVOCATION, RETURN, AND PARAMETERS

3.3.2.1General

The parameters of the CLTU-UNBIND operation shall be present in the invocation and return as specified in Table 3-3.

Table 3-3: CLTU-UNBIND Parameters

Parameter	Invocation	Return
invoker-credentials	M	
performer-credentials		M
unbind-reason	M	
result		M

3.3.2.2 **invoker-credentials**

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-UNBIND invocation (see 3.1.5).

3.3.2.3 **performer-credentials**

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of CLTU-UNBIND (see 3.1.5).

3.3.2.4 **unbind-reason**

The **unbind-reason** parameter shall specify why the CLTU-UNBIND operation is being invoked and shall contain one of the following values:

- a) 'end'—the initiator has completed the transfer of its data and is releasing the association normally: the provider may end the service instance and release its resources allocated to this service instance;

NOTE – Further invocation of the CLTU-BIND operation is not permitted even if the service instance provision period has not expired, since the service provider may release the resources allocated that service instance.

- b) 'suspend'—the initiator is suspending this service usage for an unspecified period of time: the initiator may attempt to re-bind to the responder to continue data transfer at any time prior to the end of the provision period scheduled for this service instance;
- c) 'version not supported'—the initiator does not support the version of the CLTU service proposed by the responder in the return from CLTU-BIND: this value of **unbind-reason** shall be used only if the CLTU-UNBIND is the first operation invoked following the CLTU-BIND;
- d) 'other'—the reason for the release will have to be found by other means.

3.3.2.5 **result**

3.3.2.55.1 The **result** parameter shall specify the result of the CLTU-UNBIND invocation, and shall always contain the following value: 'positive result'—the CLTU-UNBIND invocation is accepted by the responder and the association is released.

NOTE – The **result** parameter is returned for the CLTU-UNBIND operation even though the only permitted value is 'positive result', for consistency with other confirmed operations.

3.3.3 **EFFECTS**

The CLTU-UNBIND operation shall have the following effects:

- a) the association between the initiator and the responder shall be released, and the initiator and the responder shall cease to communicate with each other;
- b) the provider shall transition to state 1 ('unbound');
- c) the provider shall stop production, except for a CLTU in the process of being radiated.
- d) if `unbind-reason` is 'end', the provider may terminate the service instance and release its resources;
- e) if `unbind-reason` is not 'end', the initiator may attempt to re-bind at any time prior to the end of the provision period scheduled for this service instance.

NOTE – The provider must perform the CLTU-UNBIND operation with 'positive result' as long as it is invoked by the user while the provider is in state 2 ('ready').

3.4CLTU-START

3.4.1PURPOSE

3.4.1.1 The user shall invoke the CLTU-START operation to request that the Forward CLTU service provider prepare to receive CLTU-TRANSFER-DATA invocations (see 3.6) and enable the start of production (i.e., radiation of CLTUs).

3.4.1.2 The Forward CLTU service provider shall confirm the CLTU-START operation.

3.4.1.3 The CLTU-START operation shall allow the Forward CLTU service provider to return, to the user, the times scheduled for start and stop of production.

3.4.1.4 CLTU-START is valid only in state 2 ('ready') and shall be invoked only by the user.

3.4.2INVOCATION, RETURN AND PARAMETERS

3.4.2.1General

The parameters of the CLTU-START operation shall be present in the invocation and return as specified in Table 3-4.

Table 3-4: CLTU-START PARAMETERS

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
first-cltu-identification	M	
start-production-time		C
stop-production-time		C
result		M
diagnostic		C

3.4.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-START invocation (see 3.1.5).

3.4.2.3performer-credentials

The **performer-credentials** parameter shall contain information that enables the invoker to authenticate the return from the performance of CLTU-START (see 3.1.5).

3.4.2.4 invoke-ID

The CLTU service provider shall return unchanged the user-supplied value of the **invoke-ID** parameter (see 3.1.5).

3.4.2.5 first-cltu-identification

3.4.2.5.1 The **first-cltu-identification** parameter shall contain the cltu-identification the Forward CLTU service provider should expect to receive in the first CLTU-TRANSFER-DATA operation after the CLTU-START.

3.4.2.5.2 If the value of **first-cltu-identification** is set to 'null', the Forward CLTU service provider shall accept any first value of cltu-identification.

3.4.2.6 start-production-time

3.4.2.6.1 The **start-production-time** parameter shall be returned when the value of the **result** parameter is 'positive result'.

3.4.2.6.2 If the Forward CLTU service provider has not yet completed the PLOP:

- a) **start-time** shall be the estimated time at which the Forward CLTU service provider will be able to start radiating CLTUs;
- b) in this case **start-time** shall be determined from the schedule, with allowance for completion of the uplink acquisition sequence.

3.4.2.6.3 If the Forward CLTU service provider has already completed the PLOP and is ready to start radiating CLTUs, **start-time** shall contain the time the production status changed to 'operational'.

3.4.2.6.4 If there is a subsequent CLTU-START operation within a service instance, the **start-time** parameter shall contain the latest time the Forward CLTU service provider became ready to radiate; i.e., the most recent time the production status changed to 'operational'.

3.4.2.7 stop-production-time

3.4.2.7.1 The **stop-production-time** parameter shall be returned when the value of **result** is 'positive result'.

3.4.2.7.2 **stop-time** shall be the estimated time at which the Forward CLTU service provider will stop radiating CLTUs.

3.4.2.7.3 If the estimated stop time is not available to the Forward CLTU service provider, the **stop-time** parameter shall contain a value of 'null'.

3.4.2.8result

The **result** parameter shall be used by the Forward CLTU service provider to specify the result of the CLTU-START invocation and shall contain one of the following values:

- a) 'positive result'—the user may start invoking CLTU-TRANSFER-DATA operations;
- b) 'negative result'—the CLTU-START invocation is rejected by the provider for the reason specified in the **diagnostic** parameter. The provider cannot accept CLTU-TRANSFER-DATA invocations.

3.4.2.9diagnostic

3.4.2.9.1 If **result** is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'duplicate invoke-ID' - an operation has been invoked in which the **invoke-ID** matches that of a previous outstanding operation;
- b) 'out of service'—the provider has been taken out of service for an indefinite period by management action; **production-status** is 'halted';
- c) 'unable to comply'—the provider is unable to accept data at this time because of a possibly temporary fault condition affecting the provider; **production-status** is 'interrupted';
- d) 'production time expired'—the time period scheduled for production of CLTUs is already past;
- e) 'other'—the provider cannot accept CLTU-TRANSFER-DATA operations for unspecified reasons.

3.4.2.9.2 If **result** is 'positive result', **diagnostic** shall not be present.

3.4.3EFFECTS

3.4.3.1 If **result** is 'positive result':

- a) the provider shall transition to state 3 ('active');
- b) in the 'active' state, the provider shall accept CLTU-TRANSFER-DATA invocations.

3.4.3.2 If **result** is 'negative result':

- a) the provider shall remain in state 2 ('ready');
- b) If **diagnostic** is 'unable to comply':
 - 1) the user may re-invoke the CLTU-START operation at a later time, within the constraints of the service instance provision period;
 - 2) if the provider's complex management determines that being 'unable to comply' is more than a transient problem, it shall cause the provider to invoke the CLTU-PEER-ABORT operation.

3.5CLTU-STOP

3.5.1PURPOSE

3.5.1.1 The user shall invoke the CLTU-STOP operation to inform the provider that the user is stopping the sending of CLTUs and to request that the provider stop production, i.e., stop radiation, of CLTUs already received but not yet radiated.

NOTE – Within the constraints of the service instance provision period, the user may re-enable the reception and production of CLTUs on the provider side by invoking the CLTU-START operation.

3.5.1.2 The provider shall confirm the CLTU-STOP operation.

3.5.1.3 The CLTU-STOP operation is valid only in state 3 ('active') and shall be invoked only by the user.

3.5.2INVOCATION, RETURN AND PARAMETERS

3.5.2.1General

The parameters of the CLTU-STOP operation shall be present in the invocation and return as specified in Table 3-5.

Table 3-5: CLTU-STOP Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
result		M
diagnostic		C

3.5.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-STOP invocation (see 3.1.5).

3.5.2.3performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of CLTU-STOP (see 3.1.5).

3.5.2.4 **invoke-ID**

The provider shall return unchanged the user-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.5.2.5 **result**

The **result** parameter shall specify the result of the CLTU-STOP operation and shall contain one of the following values:

- a) 'positive result'—the CLTU-STOP invocation is accepted, and the user may invoke a CLTU-UNBIND operation or a new CLTU-START operation;
- b) 'negative result'—the CLTU-STOP invocation is rejected by the provider for the reason specified in **diagnostic**.

3.5.2.6 **diagnostic**

3.5.2.6.1 If **result** is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the **invoke-ID** parameter is the same as the **invoke-ID** value of a previous, outstanding operation;
- b) 'other'—the reason for the negative result will have to be found by other means.

3.5.2.6.2 If **result** is 'positive result', **diagnostic** shall not be present.

3.5.3 **EFFECTS**

3.5.3.1 If **result** is 'positive result':

- a) the provider shall transition to state 2 ('ready') and shall no longer accept CLTU-TRANSFER-DATA invocations;
- b) for CLTUs already buffered:
 - 1) the provider shall discard any CLTUs for which processing has not yet begun;
 - 2) the provider shall not discard, but shall complete processing, a CLTU in the process of being radiated.
- c) if a report has been requested by the user for a CLTU that is already being processed, the report shall be generated upon completion of processing and sent to the user by means of a CLTU-ASYNC-NOTIFY operation.

3.5.3.2 If **result** is 'negative result', processing continues; the provider state remains unchanged; any buffered CLTUs are not discarded.

3.6CLTU-TRANSFER-DATA

3.6.1PURPOSE

3.6.1.1 The user shall invoke the CLTU-TRANSFER-DATA operation to transmit data to the provider.

3.6.1.2 The provider shall confirm the CLTU-TRANSFER-DATA operation.

3.6.1.3 The CLTU-TRANSFER-DATA operation is valid only in state 3 ('active') and shall be invoked only by the user.

3.6.2INVOCATION, RETURN AND PARAMETERS

3.6.2.1General

The parameters of the CLTU-TRANSFER-DATA operation shall be present in the invocation and return as specified in Table 3-6.

Table 3-6: CLTU-TRANSFER-DATA Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
cltu-identification	M	M
earliest-radiation-time	M	
latest-radiation-time	M	
delay-time	M	
report	M	
data	M	
cltu-buffer-available		M
result		M
diagnostic		C

3.6.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-TRANSFER-DATA invocation (see 3.1.5).

3.6.2.3performer-credentials

The **performer-credentials** parameter shall contain information that enables the invoker to authenticate the return from the performance of CLTU-TRANSFER-DATA (see 3.1.5).

3.6.2.4 **invoke-ID**

The provider shall return unchanged the user-supplied value of the **invoke-ID** parameter (see 3.1.5).

3.6.2.5 **cltu-identification**

3.6.2.5.1 The **cltu-identification** parameter shall contain a monotonically increasing sequence number placed in all CLTU-TRANSFER-DATA operations.

NOTE – The number is used to reference the CLTU in all subsequent production reports.

3.6.2.5.2 The value of **cltu-identification** returned shall be the sequence number expected in the next CLTU-TRANSFER-DATA operation:

- a) if the transfer request is accepted, the **cltu-identification** in the return shall be one greater than the value in the request;
- b) if the transfer request is rejected, the **cltu-identification** in the return shall contain the value expected by the Forward CLTU service provider.

NOTE – It is assumed that the supported range of this parameter is sufficiently large that no two CLTUs within the CLTU buffer will have the same **cltu-identification**.

3.6.2.6 **earliest-radiation-time**

3.6.2.6.1 The **earliest-radiation-time** parameter shall be used to specify the earliest time that the provider shall start processing this CLTU

3.6.2.6.2 The **earliest-radiation-time** parameter shall either be 'unspecified', (i.e., contain a value of 'null'), or shall specify the earliest Universal Time Coordinated (UTC) time at which the leading edge of the first bit of the CLTU may begin to be radiated.

3.6.2.6.3 If the **earliest-radiation-time** parameter is 'unspecified', the provider shall begin processing immediately after any delay associated with the previous CLTU has timed out, as long as the production process is running (i.e., not 'halted' or 'interrupted').

3.6.2.7 **latest-radiation-time**

3.6.2.7.1 The **latest-radiation-time** parameter shall either contain a value of 'null', or specify the latest UTC time at which the leading edge of the first bit of the CLTU may begin to be radiated.

3.6.2.7.2 If the parameter contains a value of 'null', the CLTU shall be radiated, as long as radiation occurs before the production process is halted.

3.6.2.7.3 If **latest-radiation-time** equals **earliest-radiation-time**, radiation shall occur at this time.

3.6.2.7.4 If radiation has not begun at or before **latest-radiation-time**, an ‘expired’ exception shall occur and the CLTU shall not be radiated.

3.6.2.8delay-time

3.6.2.8.1 The **delay-time** parameter shall contain the minimum radiation delay, in microseconds, between the CLTU transferred in this operation and the next CLTU.

3.6.2.8.2 The delay shall be measured from the trailing edge of the last radiated bit of this CLTU to the leading edge of the first bit of the next CLTU.

3.6.2.8.3 If **delay-time** contains a value of zero, radiation of the next CLTU may occur immediately.

3.6.2.8.4 The effect of **delay-time** varies depending on the setting of the PLOP-in-effect parameter (see 3.1.10).

- a) When the PLOP-in-effect is PLOP-1, command modulation shall be dropped between CLTUs for the period of time specified by **delay-time**. The acquisition sequence shall be transmitted prior to radiation of the next CLTU. Time for radiation of the acquisition sequence shall be in addition to the time specified by **delay-time**.
- b) When the PLOP-in-effect is PLOP-2, an idle sequence shall be radiated between CLTUs, for the period of time specified by **delay-time**. The time specified by **delay-time** shall be increased to an integral number of bit times.

3.6.2.9report

The **report** parameter shall specify whether the provider shall invoke the CLTU-ASYNC-NOTIFY operation upon completion of the radiation of the CLTU, and shall contain one of the following values:

- a) ‘produce report’—invoke a CLTU-ASYNC-NOTIFY operation upon completion of the radiation;
- b) ‘do not produce report’—do not invoke the CLTU-ASYNC-NOTIFY operation.

3.6.2.10data

The **data** parameter shall contain a CLTU for radiation to the space element of a mission.

3.6.2.11cltu-buffer-available

The **cltu-buffer-available** parameter shall specify the remaining number of octets available for buffering of CLTUs.

3.6.2.12result

The **result** parameter shall specify the result of the CLTU-TRANSFER-DATA operation and shall contain one of the following values:

- a) 'positive result'—the CLTU-TRANSFER-DATA invocation has been accepted by the provider and the data will be buffered until processing is due.
- b) 'negative result'—the CLTU-TRANSFER-DATA invocation is rejected by the provider, for the reason returned in the **diagnostic** parameter.

3.6.2.13 diagnostic

3.6.2.13.1 If **result** is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-Id' —the value of the **invoke-ID** parameter is the same as the **invoke-ID** value of a previous, outstanding operation;
- b) 'unable to process'—the provider cannot process CLTUs, for one of the following reasons:
 - 1) the provider has been taken out of service for an indefinite period by management action, i.e., **production-status** is 'halted';
 - 2) the provider cannot process CLTUs due to a fault condition which has been reported to the user via a CLTU-ASYNC-NOTIFY operation containing a **notification-type** of 'sldu expired' or 'unable to radiate'.
- c) 'unable to store'—there is not enough buffer space available to store this CLTU;
- d) 'out of sequence'—the value of the **cltu-identification** parameter is not equal to the value expected by the provider, which is one of the following:
 - 1) in the case of the first CLTU-TRANSFER-DATA operation following a CLTU-START, the value specified in the **first-cltu-identification** parameter of the CLTU-START operation;

NOTE – In the case that a 'null' value is specified in the **first-cltu-identification** parameter of the CLTU-START operation, any value of the **cltu-identification** parameter is accepted in the first subsequent CLTU-TRANSFER-DATA operation.

 - 2) otherwise, one greater than that of the value of the **cltu-identification** parameter of the last accepted CLTU-TRANSFER-DATA operation;
- e) 'inconsistent time range'—the time specified in the **earliest-radiation-time** parameter is later than the time specified in **latest-radiation-time**;
- e) 'invalid time'—the transmission window is invalid, for one of the following reasons:
 - 1) the period from **earliest-radiation-time** to **latest-radiation-time** does not overlap with the range of times for which radiation service is scheduled;
 - 2) the service agreement does not allow the user to specify a transmission window, but the **earliest-radiation-time** and/or **latest-radiation-time** parameters contain non-null values;

- f) 'late sldu'—`latest-radiation-time` is earlier than the time the CLTU-TRANSFER-DATA operation is received by the provider;
- g) 'invalid delay time'—the value of the delay-time is outside the range set by the service agreement.
- h) 'CLTU error'—the provider has performed error checks as provided in the service agreement and has determined that this CLTU is in error; for example the CLTU exceeds the maximum size allowed for this service instance.
- i) 'other'— the reason for the negative result will have to be found by other means.

3.3.2.13.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.6.3EFFECTS

3.6.3.1 If `result` is 'positive result':

- a) the provider shall buffer the CLTU until it is due to be processed;
- b) the provider shall process CLTUs in the sequence received and according to the radiation time and delay parameters;
- c) the provider shall remain in state 3 ('active').

3.6.3.2 If `result` is 'negative result':

- a) the CLTU contained in the CLTU-TRANSFER-DATA invocation shall not be buffered;
- b) the provider shall remain in state 3 ('active').

3.7CLTU-ASYNC-NOTIFY

3.7.1PURPOSE

3.7.1.1 The provider shall invoke the CLTU-ASYNC-NOTIFY operation to notify the user of an event affecting the production of the Forward CLTU service.

NOTES

1. CLTU-ASYNC-NOTIFY is an unconfirmed operation.
2. Notifications from the provider are delivered to the user asynchronously to the flow of CLTU-TRANSFER-DATA operations from the user to the provider.

3.7.1.2 The CLTU-ASYNC-NOTIFY operation is valid in states 2 ('ready') and 3 ('active') and shall be invoked only by the provider.

3.7.2INVOCATION AND PARAMETERS

3.7.2.1General

The parameters of the CLTU-ASYNC-NOTIFY operation shall be present in the invocation as specified in Table 3-7.

Table 3-7: CLTU-ASYNC-NOTIFY Parameters

Parameters	Invocation
invoker-credentials	M
notification-type	M
event-thrown-identifier	C
cltu-last-processed	M
cltu-last-OK	M
cltu-status	C
radiation-start-time	C
radiation-stop-time	C
production-status	M
uplink-status	M

3.7.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-ASYNC-NOTIFY invocation (see 3.1.5).

3.7.2.3notification-type

The **notification-type** parameter shall describe the event being notified to the user and shall contain one of the following values:

- a) 'cltu radiated'—the CLTU identified by the value of the `cltu-last-processed` parameter successfully completed radiation. This value of `notification-type` shall be used only if the value of `report` in the associated CLTU-TRANSFER-DATA invocation was 'produce report'.
- b) 'sldu expired'—radiation of the CLTU identified by the value of the `cltu-last-processed` parameter did not begin by the time specified in the `latest-radiation-time` parameter of the associated CLTU-TRANSFER-DATA invocation.

NOTE – After the 'sldu expired' event, no further CLTUs will be radiated, and further CLTU-TRANSFER-DATA invocations will be blocked, i.e., rejected with 'unable to process' diagnostic. The service user has to clear the blocking by invoking a CLTU-STOP operation, which causes all buffered CLTUs to be discarded. In order to resume the transfer and radiation of CLTUs, after confirmation of the CLTU-STOP, the user must invoke CLTU-START.

- c) 'unable to radiate'—this event notifies an interruption in production. It indicates one of the following conditions:
 - 1) the CLTU identified by the `cltu-last-processed` parameter started radiation but did not complete due to the `production-status` becoming 'interrupted'. In this case the `cltu-status` is 'interrupted';
 - 2) the provider attempted to process the CLTU identified by the `cltu-last-processed` parameter, but could not start radiation because the `production-status` was 'interrupted'. In this case the `cltu-status` is 'undefined'.

NOTES

- 1. CLTU-ASYNC-NOTIFY is not sent when the `production-status` initially becomes 'interrupted', unless a CLTU is being radiated when the interruption occurs.
 - 2. After the 'unable to radiate' event, no further CLTUs will be radiated, and further CLTU-TRANSFER-DATA invocations will be blocked, i.e., rejected with 'unable to process' diagnostic. The service user has to clear the blocking by invoking a CLTU-STOP operation, which causes all buffered CLTUs to be discarded. After the condition causing the 'unable to radiate' event is corrected, the user can resume the transfer and radiation of CLTUs by successful invocation of CLTU-START.
- d) 'production halted'—this event occurs when the production process is stopped, with 'halted' status. The production process enters the halted state by service management decision, for example in case of malfunctioning equipment.

NOTE – After the ‘production halted’ event, no further CLTUs will be radiated, and further CLTU-TRANSFER-DATA invocations will be blocked, i.e., rejected with ‘unable to process’ diagnostic. The service user has to clear the blocking by invoking a CLTU-STOP operation, which causes all buffered CLTUs to be discarded. After management action to remove the ‘halted’ production status, the user can resume the transfer and radiation of CLTUs by successful invocation of CLTU-START.

- e) ‘production operational’—this event occurs when the production process becomes ready to radiate CLTUs and `production-status` changes to ‘operational’. The production process achieves operational status after initial establishment of the uplink channel or following actions to restore production to operational status after a halt or an interruption.
- f) ‘buffer empty’—the provider has no further CLTUs buffered for this service instance; the sequence number of the last CLTU that was processed, whether successfully or unsuccessfully, is reported in the `cltu-last-processed` parameter.

NOTE – This event shall not be notified if it is the result of a CLTU-STOP operation.

- g) ‘action list completed’—all actions associated with the event triggered by a CLTU-THROW-EVENT operation, identified by the `event-thrown-identifier` parameter, are successfully completed.
- h) ‘action list not completed’—at least one of the actions of the action list associated with the event triggered by the CLTU-THROW-EVENT operation, identified in the `event-thrown-identifier` parameter, did not complete.
- i) ‘event condition evaluated to false’—a required condition was false for the event that the CLTU-THROW-EVENT operation, identified by the `event-thrown-identifier` parameter, attempted to trigger. Therefore, the actions associated with this event have not been performed.

3.7.2.4event-thrown-identification

3.7.2.4.1 If `notification-type` is ‘action list completed’, ‘action list not completed’, or ‘event condition evaluated to false’, the **event-thrown-identifier** parameter shall be present and indicate the `event-invocation-identifier` value of the CLTU-THROW-EVENT invocation that triggered the actions whose results are reported.

3.7.2.4.2 For all other notifications, this parameter shall not be present.

3.7.2.5cltu-last-processed

For all notifications, the **cltu-last-processed** parameter shall be present. Its value shall be set as follows:

- a) If the provider has processed (i.e., attempted to radiate) no CLTUs during this service instance, the value of the `cltu-last-processed` parameter shall be set to ‘null’;

- b) Otherwise, the **cltu-last-processed** parameter shall specify the sequence number of the CLTU that the provider most recently processed or attempted to process, whether the CLTU was successfully radiated or an exception occurred.

3.7.2.6 **cltu-last-OK**

For all notifications, the **cltu-last-OK** parameter shall be present. Its value shall be set as follows:

- a) If no CLTUs have been successfully processed (i.e., radiated completely) during this service instance, the value of the **cltu-last-OK** parameter shall be set to 'null';
- b) Otherwise, the **cltu-last-OK** parameter shall specify the sequence number of the CLTU that most recently was successfully radiated.

3.7.2.7 **cltu-status**

3.7.2.7.1 For all notification, if the value of the **cltu-last-processed** parameter is 'null', the **cltu-status** parameter shall not be present.

3.7.2.7.3 Whenever the value of the **cltu-last-processed** parameter is not 'null', the **cltu-status** parameter shall be present and contain one of the following values which represent the processing state of the CLTU identified by **cltu-last-processed**:

- a) 'radiated'—radiation of the CLTU completed without aborting: the CLTU is guaranteed to have been radiated nominally;
- b) 'expired'—the latest-radiation-time of the CLTU was reached before beginning radiation: the CLTU is guaranteed not to have been radiated;
- c) 'interrupted'—radiation of the CLTU was aborted when the production status changed from 'operational' to 'interrupted': status information available to the provider indicates that the CLTU started radiating, and aborted prior to nominal completion;
- d) 'radiation started'—radiation of the CLTU started but processing is not yet complete;
- e) 'undefined'—the reported CLTU did not start radiation: the CLTU was blocked due to the 'interrupted' status of the production process.

3.7.2.8 **radiation-start-time**

3.7.2.8.1 For all notifications, if the **cltu-last-processed** is 'null', the **radiation-start-time** parameter shall not be present.

3.7.2.8.2 Whenever **cltu-last-processed** is not 'null', the **radiation-start-time** parameter shall be present, and shall contain the UTC time at which radiation of the CLTU identified by **cltu-last-processed** started, subject to the following stipulations:

- a) nominally, this parameter shall contain the radiation time of the leading edge of the first radiated bit of the CLTU;
- b) if the `cltu-status` value is 'expired' or 'undefined', the value of `radiation-start-time` shall be set to 'null'.

3.7.2.9 **radiation-stop-time**

3.7.2.9.1 For all notifications, if the `cltu-last-OK` is 'null', the **radiation-stop-time** parameter shall not be present.

3.7.2.9.2 Whenever `cltu-last-OK` is not 'null', the `radiation-stop-time` parameter shall be present, and shall contain the UTC time at which radiation of the CLTU identified in `cltu-last-OK` completed: i.e., the time of the trailing edge of the last radiated bit.

3.7.2.10 **production-status**

The **production-status** parameter shall describe the current status of the production process as perceived by the user and shall contain one of the following values:

- a) 'operational'—the production process has been configured for support, has completed the acquisition sequence, and is capable of radiating CLTUs;
- b) 'configured'—equipment has been assigned to support the service instance, but the production process is not yet capable of radiating CLTUs;
- c) 'interrupted'—the production process is stopped due to a fault;
- d) 'halted'—the production process is stopped and production equipment is out of service, due to management action.

3.7.2.11 **uplink-status**

3.7.2.11.1 The **uplink-status** parameter shall report to the user the state of the forward link obtained from the CLCW present in the return telemetry stream from the spacecraft.

3.7.2.11.2 The `uplink-status` parameter shall contain one of the following values:

- a) 'uplink status not available'—no CLCWs from the spacecraft have been received by the provider;
- b) 'no rf available'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 1;
- c) 'no bit lock'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 0, and the bit that flags No Bit Lock was set to 1;
- d) 'nominal'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 0, and the bit that flags No Bit Lock was set to 0.

3.7.3EFFECTS

3.7.3.1 If `notification-type` indicates 'sldu expired', 'unable to radiate', or 'production halted':

- a) no further CLTUs shall be radiated;
- b) further CLTU-TRANSFER-DATA invocations shall be rejected; and
- c) the provider shall remain in its original state, i.e., state 2 ('ready') or state 3 ('active').

3.7.3.2 There shall be no effect for any other values of `notification-type`.

3.8CLTU-SCHEDULE-STATUS-REPORT

3.8.1PURPOSE

3.8.1.1 The user shall invoke the CLTU-SCHEDULE-STATUS-REPORT operation to request that the provider do one of the following:

- a) send a status report immediately (see 3.9)
- b) send periodic status reports (see 3.9) or
- c) stop sending periodic status reports.

3.8.1.2 The provider shall confirm the CLTU-SCHEDULE-STATUS-REPORT operation.

3.8.1.3 For periodic reporting, the user may change the reporting period by invoking another CLTU-SCHEDULE-STATUS-REPORT operation.

3.8.1.4 The CLTU-SCHEDULE-STATUS-REPORT operation is valid only in states 2 ('ready') and 3 ('active') and shall be invoked only by the user.

3.8.2INVOCATION, RETURN, AND PARAMETERS

3.8.2.1General

The parameters of the CLTU-SCHEDULE-STATUS-REPORT operation shall be present in the invocation and return as specified in Table 3-8.

Table 3-8: CLTU-SCHEDULE-STATUS-REPORT Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
report-request-type	M	
reporting-cycle	C	
result		M
diagnostic		C

3.8.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-SCHEDULE-STATUS-REPORT invocation (see 3.1.5).

3.8.2.3 **performer-credentials**

The **performer-credentials** parameter shall contain information that enables the invoker to authenticate the return from the performance of CLTU-SCHEDULE-STATUS-REPORT (see 3.1.5).

3.8.2.4 **invoke-ID**

The performer shall return unchanged the invoker-supplied value of the **invoke-ID** parameter (see 3.1.5).

3.8.2.5 **report-request-type**

3.8.2.5.1 The **report-request-type** parameter shall specify how reporting shall be done and shall contain one of the following values:

- a) 'immediately'—send a single status report immediately;
- b) 'periodically'—send a status report every **reporting-cycle** value seconds;
- c) 'stop'—do not send further status reports.

3.8.2.5.2 If **report-request-type** is 'immediately',

- a) the provider shall stop sending status reports after the immediate status report has been sent;
- b) periodic reporting may be restarted by means of another CLTU-SCHEDULE-STATUS-REPORT invocation.

3.8.2.6 **reporting-cycle**

3.8.2.6.1 If the **report-request-type** parameter value is 'periodically', then the **reporting-cycle** parameter shall be present and shall specify the requested interval between status reports in seconds.

3.8.2.6.2 If **report-request-type** parameter value is not 'periodically', the **reporting-cycle** parameter shall not be present.

3.8.2.7 **result**

The **result** parameter shall specify the result of the CLTU-SCHEDULE-STATUS-REPORT invocation and shall contain one of the following values:

- a) 'positive result'—the CLTU-SCHEDULE-STATUS-REPORT invocation is accepted, and the provider will send the requested status report(s) or stop sending periodic status reports;

- b) 'negative result'—the CLTU-SCHEDULE-STATUS-REPORT invocation is rejected by the provider for the reason specified in the **diagnostic** parameter. The previous setting for status reporting remains in effect.

3.8.2.8diagnostic

3.8.2.8.2 If **result** is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the **invoke-ID** parameter is the same as the **invoke-ID** value of a previous, outstanding operation;
- b) 'already stopped'—the provider is not currently providing periodic reports (when **report-request-type** is 'stop');
- c) 'invalid reporting cycle'—the **reporting-cycle** value is outside the range agreed upon by management arrangement;
- d) 'other'—the reason for the negative result will have to be found by other means.

3.8.2.8.2 If **result** is 'positive result', **diagnostic** shall not be present.

3.8.3EFFECTS

3.8.3.1 If **result** is 'positive result':

- a) if **report-request-type** specifies 'immediately',
 - 1) a status report shall be sent immediately;
 - 2) the sending of any previously requested periodic status reports shall cease.
- b) if **report-request-type** specifies 'periodically':
 - 1) a status report shall be sent immediately;
 - 2) subsequent status reports shall be sent at the interval specified in the **reporting-cycle** parameter.
- c) if **report-request-type** specifies 'stop', periodic reporting shall cease.

3.8.3.2 If **result** is 'negative result', the previous setting for reporting shall remain in effect.

3.8.3.3 The provider shall remain in its original state, i.e., state 2 ('ready') or state 3 ('active').

3.9CLTU-STATUS-REPORT

3.9.1PURPOSE

3.9.1.1 The provider shall invoke the CLTU-STATUS-REPORT operation to send a status report to the user.

NOTE – CLTU-STATUS-REPORT is an unconfirmed operation.

3.9.1.2 Status reports shall be sent (or not sent) in accordance with user requests conveyed by means of the CLTU-SCHEDULE-STATUS-REPORT operation (see 3.8).

3.9.1.3 The CLTU-STATUS-REPORT operation is valid only in states 2 ('ready') and 3 ('active') and shall be invoked only by the provider.

3.9.2INVOCATION AND PARAMETERS

3.9.2.1General

The parameters of the CLTU-STATUS-REPORT operation shall be present in the invocation as specified in Table 3-9.

Table 3-9: CLTU-STATUS-REPORT Parameters

Parameters	Invocation
invoker-credentials	M
cltu-last-processed	M
cltu-last-OK	M
cltu-status	C
radiation-start-time	C
radiation-stop-time	C
production-status	M
uplink-status	M
number-of-cltus-received	M
number-of-cltus-processed	M
number-of-cltus-radiated	M
cltu-buffer-available	M

3.9.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-STATUS-REPORT invocation (see 3.1.5).

3.9.2.3 **cltu-last-processed**

3.9.2.3.1 If the provider has processed (i.e., attempted to radiate) no CLTUs during this service instance, the value of the **cltu-last-processed** parameter shall be set to 'null'.

3.9.2.3.2 Otherwise, the **cltu-last-processed** parameter shall specify the sequence number of the CLTU that the provider most recently processed or attempted to process, whether the CLTU was successfully radiated or an exception occurred.

3.9.2.4 **cltu-last-OK**

3.9.2.4.1 If no CLTUs have been successfully processed (i.e., radiated completely) during this service instance, the value of the **cltu-last-OK** parameter shall be set to 'null'.

3.9.2.4.2 Otherwise, the **cltu-last-OK** parameter shall specify the sequence number of the CLTU that most recently was successfully radiated.

3.9.2.5 **cltu-status**

3.9.2.5.1 If the value of the **cltu-last-processed** parameter is 'null', the **cltu-status** parameter shall not be present.

3.9.2.5.2 Whenever the value of the **cltu-last-processed** parameter is not 'null', the **cltu-status** parameter shall be present and contain one of the following values which represent the processing state of the CLTU identified by **cltu-last-processed**:

- a) 'radiated'—radiation of the CLTU completed without aborting: the CLTU is guaranteed to have been radiated nominally;
- b) 'expired'—the latest-radiation-time of the CLTU was reached before beginning radiation: the CLTU is guaranteed not to have been radiated;
- c) 'interrupted'—radiation of the CLTU was aborted when the production status changed from 'operational' to 'interrupted': status information available to the provider indicates that the CLTU started radiating, and aborted prior to nominal completion;
- d) 'radiation started'—radiation of the CLTU has started but not completed;
- e) 'undefined'—the reported CLTU did not start radiation: it was blocked due to the 'interrupted' status of the production process.

3.9.2.6 **radiation-start-time**

3.9.2.6.1 If the **cltu-last-processed** is 'null', the **radiation-start-time** parameter shall not be present.

3.9.2.6.2 Whenever **cltu-last-processed** is not 'null', the **radiation-start-time** parameter shall be present, and shall contain the UTC time at which radiation of the

CLTU identified by `cltu-last-processed` started, subject to the following stipulations:

- a) nominally, this parameter shall contain the radiation time of the leading edge of the first radiated bit of the CLTU;
- b) if the `cltu-status` value is 'expired' or 'undefined', the value of `radiation-start-time` shall be set to 'null'

3.9.2.7 radiation-stop-time

3.9.2.7.1 If the `cltu-last-OK` is 'null', the **radiation-start-time** parameter shall not be present.

3.9.2.7.2 Whenever `cltu-last-OK` is not 'null', the `radiation-stop-time` parameter shall be present, and shall contain the UTC time at which radiation of the CLTU identified by `cltu-last-OK` completed; i.e., the time of the trailing edge of the last radiated bit.

3.9.2.8 production-status

The **production-status** parameter shall describe the current status of the production process as perceived by the user and shall contain one of the following values:

- a) 'operational'—the production process has been configured for support, has completed the acquisition sequence, and is capable of radiating CLTUs;
- b) 'configured'—equipment has been assigned to support the service instance, but the production process is not yet capable of radiating CLTUs;
- c) 'interrupted'—the production process is stopped due to a fault;
- d) 'halted'—the production process is stopped and production equipment is out of service, due to management action;.

3.9.2.9 uplink-status

3.9.2.9.1 The **uplink-status** parameter shall report to the user the state of the forward link obtained from the CLCW present in the return telemetry stream from the spacecraft.

3.9.2.9.2 The `uplink-status` parameter shall contain one of the following values:

- a) 'uplink status not available'—no CLCWs from the spacecraft have been received by the provider;
- b) 'no rf available'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 1;
- c) 'no bit lock'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 0, and the bit that flags No Bit Lock was set to 1;

- d) 'nominal'—the provider has received at least one CLCW; in the last CLCW received by the provider, the bit that flags No RF Available was set to 0, and the bit that flags No Bit Lock was set to 0.

3.9.2.10number-of-cltus-received

3.9.2.10.1 The **number-of-cltus-received** parameter shall specify the total number of CLTUs received during this service instance.

3.9.2.10.2 Only CLTUs that are checked successfully and buffered shall be counted in the total.

3.9.2.10.3 If no CLTUs have been received, this parameter shall be set to zero.

3.9.2.11number-of-cltus-processed

3.9.2.11.1 The **number-of-cltus-processed** parameter shall specify the number of CLTUs that the provider attempted to radiate during this service instance, including

- CLTUs that are determined to have expired;
- CLTUs that are determined to have aborted;
- a CLTU in the process of being radiated.

3.9.2.11.2 If no CLTUs have been processed, this parameter shall be set to zero.

3.9.2.12number-of-cltus-radiated

3.9.2.12.1 The **number-of-cltus-radiated** shall contain the count of CLTUs successfully radiated during this service instance.

3.9.2.12.2 A CLTU in the process of being radiated shall not be included in this count.

3.9.2.12.3 If no CLTUs have been radiated, this parameter shall be set to zero.

3.9.2.13cltu-buffer-available

The **cltu-buffer-available** parameter shall specify the remaining number of octets available for storage of CLTUs. Once the provider starts to process a CLTU, it shall be considered removed from the buffer.

3.9.3EFFECTS

3.9.3.1 Status information shall be delivered to the user.

3.9.3.2 The provider shall remain in its original state, i.e., state 2 ('ready') or state 3 ('active').

3.10 CLTU-GET-PARAMETER

3.10.1 PURPOSE

3.10.1.1 The user shall invoke the CLTU-GET-PARAMETER operation to ascertain the value of a CLTU service parameter.

3.10.1.2 The provider shall confirm the CLTU-GET-PARAMETER operation.

3.10.1.3 The CLTU-GET-PARAMETER operation is valid only in states 2 ('ready') and 3 ('active') and shall be invoked only by the user.

3.10.2 INVOCATION, RETURN, AND PARAMETERS

3.10.2.1 General

The parameters of the CLTU-GET-PARAMETER operation shall be present in the invocation and return as specified in Table 3-10.

Table 3-10: CLTU-GET-PARAMETER Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
cltu-parameter	M	C
parameter-value		C
result		M
diagnostic		C

3.10.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-GET-PARAMETER invocation (see 3.1.5).

3.10.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of CLTU-GET-PARAMETER (see 3.1.5).

3.10.2.4 invoke-ID

The performer shall return unchanged the invoker-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.10.2.5 cltu-parameter

3.10.2.5.1 The **cltu-parameter** parameter shall specify the CLTU parameter whose value is to be returned to the user and shall contain one of the values listed in Table 3-11.

Table 3-11: Forward CLTU Parameters

Parameter	Description
bit-lock-required	If the value is 'yes', the provider must verify from the CLCW that 'No bit lock' in the CLCW is false before setting the production status to 'operational' and radiating the first CLTU.
delivery-mode	'fwd online'
expected-cltu-identification	The expected value of the cltu-identification parameter to be received in the next CLTU-TRANSFER-DATA invocation. If no CLTU-START has been received, zero shall be returned as the default value of this parameter.
expected-event-invocation-identification	The expected value of the event-invocation-identification parameter to be received in the next CLTU-THROW-EVENT invocation. The initial value of this parameter is zero.
maximum-cltu-length	The size, in bits, of the maximum-length CLTU that will be accepted by the provider for this service instance.
modulation-frequency	The frequency of the primary modulation of the rf carrier, expressed in tenths of Hertz. Primary modulation means the subcarrier frequency, when applicable, otherwise the frequency of direct data modulation.
modulation-index	'modulation-index'—the amount of RF carrier suppression, expressed in degrees.
plop-in-effect	The physical layer operation procedure being used: PLOP-1 or PLOP-2.
reporting-cycle	'null' if cyclic reporting is off, otherwise the time (in seconds) between successive CLTU-STATUS-REPORT invocations (see 3.8.2.6).
return-timeout-period	Maximum time period (in seconds) permitted between the invocation of a Forward CLTU operation and the receipt by the invoker of the return (for confirmed operations). (See 4.1.3).
rf-available-required	If the value is 'yes', the provider must verify from the CLCW that 'No RF available' in the CLCW is false before radiating Acquisition Sequence and, when the bit-lock-required parameter is 'no', also setting the production status to 'operational'.
subcarrier-to-bit-rate-ratio	When subcarrier modulation is used, the value represents the ratio of the subcarrier frequency to the uplink data rate (i.e., the bit rate). The value is set to 1 when data is directly modulated onto the carrier.

3.10.2.5.2 `cltu-parameter` is conditionally present in the return based on the `result` parameter:

- a) if the value of `result` is 'positive result', `cltu-parameter` shall be present in the return;
- b) if the value of `result` is 'negative result', `cltu-parameter` shall not be present in the return.

3.10.2.6 parameter-value

3.10.2.6.1 The **parameter-value** parameter shall contain the value for the parameter specified by `cltu-parameter` as described in 3.10.2.5.

3.10.2.6.2 `parameter-value` is conditionally present in the return based on the `result` parameter:

- a) if the value of `result` is 'positive result', `parameter-value` shall be present;
- b) if the value of `result` is 'negative result', `parameter-value` shall not be present.

3.10.2.6result

The **result** parameter shall specify the result of the CLTU-GET-PARAMETER invocation and shall contain one of the following values:

- a) 'positive result'—the CLTU-GET-PARAMETER invocation is accepted by the provider, and the value of the specified CLTU parameter is provided in the positive return;
- b) 'negative result'—the CLTU-GET-PARAMETER invocation is rejected by the provider for the reason specified in the `diagnostic` parameter.

3.10.2.8 diagnostic

3.10.2.8.1 If `result` is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the `invoke-ID` parameter is the same as the `invoke-ID` value of a previous, outstanding operation;
- b) 'unknown parameter'—the parameter specified by the value of `cltu-parameter` is not known to the service provider;
- c) 'other'—the reason for the negative result will have to be found by other means.

3.10.2.8.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.10.3EFFECTS

3.10.3.1 If `result` is 'positive result', the value of the parameter specified by the value of the `cltu-parameter` value shall be returned to the user in the positive return.

3.10.3.2 If `result` is 'negative result', the value of the parameter specified by the value of the `cltu-parameter` parameter shall not be returned to the user.

3.10.3.3 The provider shall remain in its original state shall, i.e., state 2 ('ready') or state 3 ('active').

3.11CLTU-THROW-EVENT

3.11.1PURPOSE

3.11.1.1 The user shall invoke the CLTU-THROW-EVENT operation in order to cause the provider to forward to Complex Management an event that requires management action.

NOTE – An example of such an event is change of the uplink bit rate.

3.11.1.2 The performer shall confirm the CLTU-THROW-EVENT operation.

NOTE – The return will only show whether the invocation itself has been accepted or rejected, but not if the actions associated with the event have been performed successfully. The provider will invoke a CLTU-ASYNC-NOTIFY operation to inform the user about the event.

3.11.1.3 The CLTU-THROW-EVENT operation is valid in states 2 ('ready') and 3 ('active') and shall be invoked only by the user.

3.11.2INVOCATION, RETURN, AND PARAMETERS

3.11.2.1General

The parameters of the CLTU-THROW-EVENT operation shall be present in the invocation and return as specified in Table 3-12.

Table 3-12: CLTU-THROW-EVENT Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
event-identification	M	
event-invocation-identifier	M	M
result		M
diagnostic		C

3.11.2.2invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the CLTU-THROW-EVENT invocation (see 3.1.5).

3.11.2.3performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of CLTU-THROW-EVENT (see 3.1.5).

3.11.2.4 **invoke-ID**

The performer shall return unchanged the invoker-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.11.2.5 **event-identification**

The **event-identification** parameter shall identify the event to be forwarded to Complex Management by the service provider.

3.11.2.6 **event-invocation-identifier**

3.11.2.6.1 The **event-invocation-identifier** parameter shall contain a monotonically increasing sequence number and shall be

- a) set to zero for the first CLTU-THROW-EVENT invocation following the first successful CLTU-BIND to the service instance;
- b) incremented for each CLTU-THROW-EVENT invocation accepted by the provider.

3.11.2.6.2 The provider shall set the value of the **event-invocation-identifier** parameter to the sequence number expected in the next CLTU-THROW-EVENT invocation:

- a) if the invocation is accepted, the value of **event-invocation-identifier** in the return shall be one greater than the value in the invocation parameter;
- b) if the invocation is rejected, **event-invocation-identifier** in the return shall contain the value expected by the provider.

3.11.2.6.3 The user shall set the value of **event-invocation-identifier** in the next CLTU-THROW-EVENT invocation to the value returned by the provider.

3.11.2.6.4 The value expected by the provider shall not be altered when the association between user and provider is reestablished after being released or aborted.

3.11.2.6.5 The provider shall use the user-supplied value of **event-invocation-identifier** to reference the respective event in all subsequent reports.

3.11.2.7 **result**

The **result** parameter shall specify the result of the CLTU-THROW-EVENT invocation and shall contain one of the following values:

- a) ‘positive result’—the CLTU-THROW-EVENT invocation is accepted by the provider and will be forwarded to Complex Management;
- b) ‘negative result’—the CLTU-THROW-EVENT invocation is rejected by the provider for the reason specified in the **diagnostic** parameter. The provider did not forward the event.

3.11.2.8 diagnostic

3.11.2.8.1 If `result` is 'negative result', `diagnostic` shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the `invoke-ID` parameter is the same as the `invoke-ID` value of a previous, outstanding operation;
- b) 'operation not supported'—the CLTU-THROW-EVENT operation is not supported by the provider;
- c) 'out of sequence'—the value of the `event-invocation-identifier` parameter is not equal to the value expected by the provider;
- d) 'no such event'—the `event-identification` specified by the user through this CLTU-THROW-EVENT invocation is not in the set of `event-identifications` recognized by Complex Management;
- e) 'other'—the reason for the negative result will have to be found by other means.

3.11.2.8.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.11.3 EFFECTS

3.11.3.1 If `result` is 'positive result':

- a) the event shall be forwarded to Complex Management;
- b) the CLTU provider shall pass notifications received from the event handler to the user by invoking the CLTU-ASYNC-NOTIFY operation.

NOTE – The effects of the mission-specific actions performed depend on the event handler and are outside the scope of this Recommendation.

3.11.3.2 If `result` is 'negative result', the event shall not be forwarded to Complex Management.

3.11.3.3 The provider shall remain in its original state shall, i.e., state 2 ('ready') or state 3 ('active').

3.12CLTU-PEER-ABORT

3.12.1PURPOSE

3.12.1.1 The user or provider shall invoke the CLTU-PEER-ABORT operation to notify the peer system that the local application detected an error that requires that the association between them be terminated.

NOTE – CLTU-PEER-ABORT is an unconfirmed operation.

3.12.1.2 The CLTU-PEER-ABORT operation is valid only in states 2 ('ready') and 3 ('active') and may be invoked by either the user or the provider.

3.12.2INVOCATION AND PARAMETERS

3.12.2.1General

The parameters of the CLTU-PEER-ABORT operation shall be present in the invocation as specified in **Table 3-13**.

Table 3-13: CLTU-PEER-ABORT Parameters

Parameters	Invocation
diagnostic	M

3.12.2.2diagnostic

The **diagnostic** parameter shall specify why the CLTU-PEER-ABORT is being issued and shall contain one of the following values:

- a) 'access denied'—a responder with an identity as presented in the **responder-identifier** parameter of the CLTU-BIND return is not registered at the initiator.
- b) 'unexpected responder ID'—the value of the **responder-identifier** parameter in the CLTU-BIND return did not match the identity the initiator intended to bind to.
- c) 'operational requirement'—the local system had to abort the association to accommodate some other operational need;
- d) 'protocol error'—the local application detected an error in the sequencing of CLTU operations;
- e) 'communications failure'—the communications service on the other side of the gateway was disrupted.

NOTE – This diagnostic value is only applicable when the SLE applications are communicating via a gateway (see 2.6.4.5).

- f) 'encoding error'—the local application detected an error in the encoding of one or more operation parameters, or did not recognize the operation;

- g) 'return timeout'—the local application detected that a return of a confirmed operation was not received within the time specified by the return-timeout-period parameter (see 3.1.10);
- h) 'end of service instance provision period'—the local application detected that the period scheduled for provision of service for this service instance has ended, and the initiator has not invoked an UNBIND operation;
- i) 'unsolicited invoke-ID'—the local application received a return with an invoke-ID that does not match with the invoke-ID of any of the operations for which the return is pending;
- j) 'other'—the local application detected an unspecified error during the processing of one or more operations.

NOTE – CLTU-PEER-ABORT does not carry an `invoker-credentials` parameter. Although it is conceivable that an intruder might use the CLTU-PEER-ABORT operation for a denial-of-service attack, this would require that the intruder succeed in breaking into an existing connection. If an intruder has this capability, then a denial-of-service attack can be much more easily accomplished by disrupting communications at a layer lower than the application layer. Therefore, authentication of the CLTU-PEER-ABORT operation would not provide improved protection against such attacks.

3.12.3EFFECTS

3.12.3.1 The association shall be aborted and the provider shall transition to state 1 ('unbound').

NOTE – The CLTU-PEER-ABORT operation shall be truly abortive. This implies that the transfer of the SLE-PDU carrying this invocation shall not be delayed by other invocations submitted earlier and possibly being buffered by the underlying communication service. This needs to be considered when the mapping onto the underlying communication services is decided.

3.12.3.2 All CLTUs buffered for this service instance by the provider shall be discarded.

3.12.3.3 The values of all service provision parameters shall return to the initial values set by service management, including the values of any parameters changed by the service user.

3.12.3.4 The provider shall maintain accumulated statistics such as are reported in CLTU-STATUS-REPORT for the duration of the service instance provision period.

3.12.3.5 Parameters that serve to relate notifications on an activity to operations that triggered that activity shall not be altered when the association is aborted.

4CLTU PROTOCOL

4.1GENERIC PROTOCOL CHARACTERISTICS

This section specifies the handling of invalid SLE-PDUs and other failures affecting the protocol. .

4.1.1UNEXPECTED PROTOCOL DATA UNIT

If the peer application sends an invocation or return not allowed in the current state of the performer, the performer shall abort the association by invoking the CLTU-PEER-ABORT operation containing a 'protocol error' diagnostic.

4.1.2INVALID OR UNDECODABLE PROTOCOL DATA UNIT

If the application receives an invocation or return that contains an unrecognized operation type, contains a parameter of a wrong type, or is otherwise not decodable, the association shall be aborted by means of the CLTU-PEER-ABORT operation containing an 'encoding error' diagnostic.

4.1.3MISSING RETURN

For confirmed operations:

- a) If the invoker does not receive the return from the performer within the return-timeout-period specified by service management, the invoker shall abort the association by invoking the CLTU-PEER-ABORT operation containing a 'return time-out' diagnostic;
- b) the return-timeout-period shall be chosen taking into account performance of user and provider applications as well as the delays introduced by the underlying communications service;

NOTES

- 1 Reasonably short return-timeout-periods imply that the generation of the return from an operation must not depend on any human interaction.
- 2 After invoking the CLTU-UNBIND operation, the initiator must not invoke any further operations nor send any returns. The responder is not required to send any pending returns after having received the CLTU-UNBIND invocation. Therefore, following a CLTU-UNBIND invocation, the 'missing return' event may occur.

4.1.4UNSOLICITED RETURN

If the application receives a return with an `invoke-ID` parameter value that does not correspond to any invocation for which the return is still pending, the association shall be aborted by means of the CLTU-PEER-ABORT operation containing an 'unsolicited invoke-ID' diagnostic.

4.1.5COMMUNICATIONS FAILURE

4.1.5.1 If an SLE entity, i.e., an SLE user or provider, is informed by its underlying communications service that the connection with the peer entity has been closed due to a

communications failure, then this entity shall consider that the association with the peer entity has been aborted.

4.1.5.2 Occurrence of the above described communications failure event shall be referred to as a ‘protocol abort’.

NOTE – The exact criteria for determining when communications have failed and the communications connection is closed depend on the characteristics of the communications technology used and may be specific to a given implementation. However, it is a firm requirement on any communication service underlying an SLE transfer service that it includes suitable means to ascertain the status of the connection (see 1.3.1). In the case of communications protocols that do not intrinsically provide such a capability, the transmission of a periodic “heartbeat” indicator or equivalent may need to be implemented.

4.1.5.3 The SLE entity shall be able to accept additional diagnostic information that the communications service provides together with the notification of the ‘protocol abort’ event.

NOTE – Such diagnostic information is assumed to be descriptive of the specific nature or cause of the fault. Its type and format will be implementation dependent. The SLE entity may log this information, display it to the user, etc.

4.1.5.4 Following a ‘protocol abort’ event:

- a) the CLTU provider shall transition to state 1 (‘unbound’) and discard all CLTUs buffered for this service instance;
- b) the values of all service provision parameters shall return to the initial values set by service management, including the values of any parameters changed by the service user;
- c) neither the SLE user nor the SLE provider shall attempt further communications except that the user may reinvoke the CLTU-BIND operation.

4.1.5.5 Statistical information required for the generation of the status report shall be retained during the service instance provision period, regardless of a ‘protocol abort’ event.

4.1.5.6 Parameters that serve to relate notifications on an activity to operations that triggered that activity (such as `cltu-identifier`) shall not be altered in the event of a ‘protocol abort’.

4.1.6 ACCESS CONTROL

4.1.6.1 The initiator of an association shall present its own identity in the `initiator-identifier` parameter of the CLTU-BIND invocation.

4.1.6.2 If the initiator is not a user registered at the responder, the responder shall deny access to this initiator by sending a negative CLTU-BIND return with `diagnostic` set to ‘access denied’ and `performer-credentials` set to ‘unused’.

4.1.6.3 If the initiator receives an CLTU-BIND return with a `responder-identifier` value that does not correspond to the identity of any responder registered at this initiator, the initiator shall abort the association by means of the CLTU-PEER-ABORT operation containing an ‘access denied’ `diagnostic`.

4.1.6.4 If the initiator is registered at the responder, the responder shall attempt to authenticate the CLTU-BIND invocation (see 3.1.5) as required for the given initiator. If authentication succeeds, but the initiator is not authorized for the service instance indicated in the `service-instance-identifier` parameter of the CLTU-BIND invocation, the responder shall generate an CLTU-BIND return with `result` set to 'negative result' and `diagnostic` set to 'service instance not accessible to this initiator'.

4.1.6.5 If the responder is registered at the initiator, the initiator shall attempt to authenticate the CLTU-BIND return (see 3.1.5) as required for the given responder. If authentication succeeds, but the `responder-identifier` value is different from the expected responder identity, the initiator shall abort the association by means of the CLTU-PEER-ABORT operation containing an 'unexpected responder ID' `diagnostic`.

NOTE – In case authentication fails, initiator and responder shall behave as specified in 4.1.7.

4.1.7 FAILING AUTHENTICATION

4.1.7.1 An incoming invocation or return shall be ignored if the credentials parameter cannot be authenticated when, by management arrangement, credentials are required.

4.1.7.2 If an invocation is ignored, the operation shall neither be confirmed nor performed.

4.1.7.3 If a return is ignored, it shall be as if no report of the outcome of the operation has been received.

4.2 STATE TABLE

4.2.1 SCOPE OF THE STATE TABLE

The state table specified in 4.2.2 represents one instance of service, and thus one association. Once the association is established, if a CLTU-BIND invocation is received, it shall be rejected with a CLTU-BIND return where the `result` parameter is 'negative result' and the `diagnostic` parameter is 'already bound'. This event shall not affect the association already in place.

4.2.2 STATE TABLE

NOTE – The state table specifies service operation interactions and state transitions for the provider system. The structure of the state table is:

- The second column lists all incoming events. Where these events correspond to the arrival of an incoming SLE-PDU, the ASN.1 type defined for this SLE-PDU in Annex A is indicated in parentheses ().
- Where an event is internal to the provider, its description is put in quotation marks: ' '. These events are defined in Table 4-2.
- The three columns (one column per state) on the right side of the table specify the behavior the provider will exhibit depending on the current state and the incoming event. In some cases, the behavior in addition depends on Boolean

conditions, also referred to as predicates. Such predicates are put in double quotes “ ”. The predicates are defined in Table 4-3. Predicates that are simple Boolean variables set only by the state machine itself, are referred to as Boolean Flags and specified in Table 4-4. The dependency on a predicate is presented in the form of an ‘IF <condition> THEN <action> [ELSEIF <condition> THEN <action>] ELSE <action>’ clause.

- If the action is simply to send a specific SLE-PDU, this is indicated by the ASN.1 type of this SLE-PDU in parentheses (). The convention has been introduced to use ‘+’ and ‘-’ to represent positive and negative returns, respectively. Where several actions are to be taken, this is indicated by a ‘compound action’ which is put in curly braces { }. The individual actions making up this compound action are identified in Table 4-5.
- ‘Not applicable’ is stated where the given event can only occur in the given state because of an implementation error on the provider side.
- Where the consequences of an incoming event are not visible to the user because the provider does not send any SLE-PDU in reaction to the given event, the action is put in square brackets [].
- State transitions are indicated by an arrow and the number of the state that will be entered; for example, → 1 indicates the transition to state 1.
- The actions to be taken and the state transition are considered one atomic operation. The sequence is irrelevant except that SLE-PDUs shall be sent in the sequence stated in the table.

CCSDS HISTORICAL DOCUMENT
DRAFT CCSDS RECOMMENDATION FOR SLE FORWARD CLTU SERVICE

Table 4-1: Behavior of Provider

	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
1.	(cltuBindInvocation)	IF "positive result" THEN (+cltuBindReturn) → 2 ELSE (-cltuBindReturn)	IF "same service instance" THEN (-cltuBindReturn ('already bound')) ELSE {peer abort ('protocol error')} → 1	IF "same service instance" THEN (-cltuBindReturn ('already bound')) ELSE {peer abort ('protocol error')} → 1
2.	'end of service instance provision period'	[ignore]	{peer abort ('end of service instance provision period')} → 1	{peer abort ('end of service instance provision period')} → 1
3.	(cltuUnbindInvocation)	[ignore]	(cltuUnbindReturn) → 1 stop reporting-cycle timer IF "end" THEN release resources ELSE [ignore]	{peer abort ('protocol error')} → 1
4.	(cltuStartInvocation)	[ignore]	IF "positive result" THEN (+cltuStartReturn) → 3 ELSE (-cltuStartReturn)	{peer abort ('protocol error')} → 1
5.	(cltuStopInvocation)	[ignore]	{peer abort ('protocol error')} → 1	IF "positive result" THEN {initiate stop} → 2 ELSE (-cltuStopReturn)
6.	(cltuTransferDataInvocation)	[ignore]	{peer abort ('protocol error')} → 1	IF "positive result" .AND. (.NOT. "service instance blocked") THEN buffer CLTU (+cltuTransferDataReturn) ELSE discard CLTU (-cltuTransferDataReturn)

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	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
7.	(cltuScheduleStatusReportInvocation)	[ignore]	IF "positive result" THEN (+cltuScheduleStatusReportReturn) IF "immediately" THEN {immediate report} ELSE IF "periodically" THEN {periodic report} ELSE stop reporting-cycle timer ELSE (-cltuScheduleStatusReportReturn)	IF "positive result" THEN (+cltuScheduleStatusReportReturn) IF "immediately" THEN {immediate report} ELSE IF "periodically" THEN {periodic report} ELSE stop reporting-cycle timer ELSE (-cltuScheduleStatusReportReturn)
8.	'reporting-cycle timer expired'	Not applicable	{periodic report}	{periodic report}
9.	(cltuGetParameterInvocation)	[ignore]	IF "positive result" THEN (+cltuGetParameterReturn) ELSE (-cltuGetParameterReturn)	IF "positive result" THEN (+cltuGetParameterReturn) ELSE (-cltuGetParameterReturn)
10.	(cltuThrowEventInvocation)	[ignore]	IF "positive result" THEN (+cltuThrowEventReturn) forward event to Complex Management ELSE (-cltuThrowEventReturn)	IF "positive result" THEN (+cltuThrowEventReturn) forward event to Complex Management ELSE (-cltuThrowEventReturn)
11.	'cltu radiated'	Not applicable	IF "report" THEN {notify} ELSE [ignore]	IF "report" THEN {notify} ELSE [ignore]
12.	'sldu expired'	Not applicable	Not applicable	{notify and block}
13.	'unable to radiate'	Not applicable	Not applicable	{notify and block}
14.	'production halted'	Not applicable	{notify}	{notify and block}
15.	'production operational'	Not applicable	{notify}	{notify}
16.	'buffer empty'	Not applicable	{notify}	{notify}
17.	'action list completed'	Not applicable	{notify}	{notify}
18.	'action list not completed'	Not applicable	{notify}	{notify}
19.	'event condition evaluated to false'	Not applicable	{notify}	{notify}
20.	(cltuPeerAbortInvocation)	[ignore]	{clean up} → 1	{clean up} → 1

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	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
21.	'protocol abort'	[ignore]	{clean up} → 1	{clean up} → 1
22.	'unsolicited invoke-ID'	[ignore]	{peer abort ('unsolicited invoke-ID')} → 1	{peer abort ('unsolicited invoke-ID')} → 1
23.	'invalid SLE-PDU'	[ignore]	{peer abort ('encoding error')} → 1	{peer abort ('encoding error')} → 1
24.	'not authenticated SLE-PDU'	[ignore]	[ignore]	[ignore]

Table 4-2: Event Description References

Event	Reference
'action list completed'	3.7.2.3
'action list not completed'	3.7.2.3
'buffer empty'	3.7.2.3
'cltu radiated'	3.7.2.3
'end of service instance provision period'	3.12.2.2
'event condition evaluated to false'	3.7.2.3
'invalid SLE-PDU'	4.1.2
'not authenticated SLE-PDU'	4.1.7
'production halted'	3.7.2.3
'production operational'	3.7.2.3
'protocol abort'	4.1.5
'reporting-cycle timer expired'	3.8.2
'sldu expired'	3.7.2.3
'unable to radiate'	3.7.2.3
'unsolicited invoke-ID'	3.12.2.2

Table 4-3: Predicate Definitions

Predicate	Evaluates to TRUE if
"end"	all checks on the (cltuUnbindInvocation) PDU are passed and the <code>unbind-reason</code> parameter value is 'end'
"immediately"	all parameter checks on the (cltuScheduleStatusReportInvocation) PDU are passed and the <code>report-request-type</code> parameter value is 'immediately'
"periodically"	all parameter checks on the (cltuScheduleStatusReportInvocation) PDU are passed and the <code>report-request-type</code> parameter value is 'periodically'
"positive result"	all checks on the given invocation PDU are passed
"same service instance"	the <code>service-instance-identifier</code> value in the CLTU-BIND invocation denotes a service instance that is already bound.

Table 4-4: Boolean Flags

Flag Name	Initial Value
"service instance blocked"	FALSE

Table 4-5: Compound Action Definitions

Name	Actions Performed
{clean up}	a) stop reporting-cycle timer b) clear CLTU buffer c) reset service provision parameters
{immediate report}	a) (cltuStatusReportInvocation) b) stop reporting-cycle timer
{initiate stop}	a) clear CLTU buffer b) set "service instance blocked" to FALSE c) (+cltuStopReturn)
{notify}	(cltuAsyncNotifyInvocation) with notification-type set to the value corresponding to the incoming event
{notify and block}	a) (cltuAsyncNotifyInvocation) with notification-type set to the value corresponding to the incoming event b) set "service instance blocked" to TRUE
{peer abort ('xxxx')}	a) (cltuPeerAbortInvocation) with diagnostic set to 'xxxx' b) stop reporting-cycle timer c) clear CLTU buffer
{periodic report}	a) (cltuStatusReportInvocation) b) set reporting-cycle timer to the reporting-cycle value in the most recent (cltuScheduleStatusReportInvocation) c) start reporting-cycle timer

ANNEX A**ASN.1 PROTOCOL SPECIFICATION
FORWARD CLTU SERVICE
DATA TYPE DEFINITIONS**

(This annex is part of the Recommendation)

A.1 INTRODUCTION

This annex defines the data types that are used by the Forward CLTU service. It is intended to provide a clear specification of these data types and to avoid ambiguity. It is not intended to constrain how these data types are implemented or encoded. These definitions are suitable for inclusion in any type of ASN.1 based protocol that implements the Forward CLTU service.

The data type definitions are presented in six ASN.1 modules.

Subsection A.2.1 contains basic data types that are to be common with other SLE transfer services. As additional services are specified by CCSDS, further data types may be added to this module or existing types may be extended. This does, however, not invalidate the module in its present form because an implementation compliant with a future extended version of this module is still fully interoperable with an implementation based on its present version.

Subsection A.2.2 specifies the SLE-PDUs exchanged between an SLE user and an SLE provider application in order to establish, release or abort an association. They are common among SLE transfer service types.

Subsection A.2.3 specifies SLE-PDUs related to invocations and returns that are common to SLE transfer service types.

Subsection A.2.4 specifies data types specific to the Forward CLTU service. In part, these specific types are derived from types specified in A.2.1 by means of subtyping.

Subsection A.2.5 specifies all incoming (from a provider point of view) SLE-PDUs. Where applicable, these SLE-PDUs are mapped to the generic SLE-PDUs defined in A.2.2 and A.2.3.

Subsection A.2.6 specifies in the same way all outgoing SLE-PDUs.

Although subsections A.2.2, A.2.3, A.2.5 and A.2.6 define the SLE-PDUs that will be exchanged between the SLE provider and user applications, they shall not be interpreted as requiring that these PDUs shall be completely mapped to the user data field of the underlying communications protocol. Depending on the communications protocol(s) used, part of the SLE-PDUs may be used to determine the appropriate setting of protocol control information.

A.2 CLTU DATA TYPE SPECIFICATION**A.2.1 SLE TRANSFER SERVICE COMMON TYPES**

CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS    ConditionalTime
           ,
           , Credentials
           , DeliveryMode
           , Diagnostics
           , Duration
           , ForwardDuStatus
           , IntPosLong
           , IntPosShort
           , IntUnsignedLong
           , IntUnsignedShort
           , InvokeId
           , ParameterName
           , SlduStatusNotification
           , SpaceLinkDataUnit
           , Time
           ;

```

```

ConditionalTime ::= CHOICE
{
  undefined [0] NULL
  known     [1] Time
}

```

```

-- If credentials are used, it will be necessary that
-- the internal structure of the octet string is known
-- to both parties. Since the structure will depend on
-- algorithm used, it is not specified here. However,
-- the peer entities may use ASN.1 encoding to make the
-- internal structure visible.

```

```

Credentials ::= CHOICE
{
  unused [0] NULL
  used   [1] OCTET STRING (SIZE (8 .. 256))
}

```

```

DeliveryMode ::= INTEGER
{
  rtnTimelyOnline (0)
  , rtnCompleteOnline (1)
  , rtnOffline (2)
  , fwdOnline (3)
  , fwdOffline (4)
}

```

```

Diagnostics ::= INTEGER
{
  duplicateInvokeId (100)
  , otherReason (127)
}

```

```

-- The Duration is expressed in microseconds
Duration ::= IntUnsignedLong

```

```

ForwardDuStatus          ::= INTEGER
{
  radiated                (0)
, expired                 (1)
, interrupted             (2)
, transferred             (3) -- FSP
, productionStarted       (4) -- CLTU: 'radiation started'
, undefined               (5) -- FSP and CLTU
, unsupportedTransmissionMode (6) -- FSP
}

-- 1 to (2^32)-1
IntPosLong               ::= INTEGER (1 .. 4294967295)

-- 1 to (2^16)-1
IntPosShort              ::= INTEGER (1 .. 65535)

-- 0 to (2^32)-1
IntUnsignedLong          ::= INTEGER (0 .. 4294967295)

-- 0 to (2^16)-1
IntUnsignedShort         ::= INTEGER (0 .. 65535)

InvokeId                 ::= IntUnsignedShort

ParameterName            ::= INTEGER
{
  aggregationTimeoutPeriod (0)
, aggregationUsage         (1)
, apidList                 (2)
, bitLockRequired          (3)
, bufferSize               (4)
, copInEffect              (5)
, deliveryMode             (6)
, directiveInvocation       (7)
, expectedDirectiveIdentification (8)
, expectedEventInvocationIdentification (9)
, expectedSlduIdentification (10)
, fecfUsage                (11)
, fopSlidingWindow         (12)
, fopState                 (13)
, idlePattern              (14)
, latencyLimit             (15)
, mapList                  (16)
, mapMuxControl            (17)
, mapMuxScheme             (18)
, maximumFrameLength       (19)
, maximumPacketLength      (20)
, maximumSlduLength        (21)
, modulationFrequency      (22)
, modulationIndex          (23)
, permittedGvcidSet        (24)
, plopInEffect             (25)
, reportingCycle           (26)
, requestedFrameQuality    (27)
, requestedGvcid           (28)
, returnTimeoutPeriod      (29)
, rfAvailable              (30)
, rfAvailableRequired      (31)
, segmentHeader            (32)
, sentQueueSize            (33)
, subcarrierToBitRateRatio (34)
, timeoutType              (35)
, timerInitial             (36)
, transmissionLimit        (37)
, vcMuxControl             (38)
}

```

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```
, vcMuxScheme                (39)
, virtualChannel              (40)
}

SlduStatusNotification ::= INTEGER
{ produceNotification      (0)
, doNotProduceNotification (1)
}

SpaceLinkDataUnit ::= OCTET STRING (SIZE (7 .. 65536))

Time ::= CHOICE
{ ccsdsFormat      [0] TimeCCSDS
}

TimeCCSDS ::= OCTET STRING (SIZE(8))
-- P-field is implicit (not present, defaulted to 41 hex
-- T-field:
-- 2 octets: number of days since 1958/01/01 00:00:00
-- 4 octets: number of milliseconds of the day
-- 2 octets: number of microseconds of the millisecond
--          (set to 0 if not used)
-- This definition reflects exactly the format of the CCSDS defined
-- time tag as used in spacelink data units.

END
```

A.2.2 SLE-Transfer Service BIND Types

CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```
EXPORTS  SleBindInvocation
,        SleBindReturn
,        SlePeerAbort
,        SleUnbindInvocation
,        SleUnbindReturn
;
```

```
IMPORTS  Credentials
,        IntPosShort
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES
;
```

```
-- =====
-- The first part of the module definition contains the SLE-PDUs
-- =====
```

```
SleBindInvocation      ::= SEQUENCE
{ invokerCredentials    Credentials
, initiatorIdentifier    AuthorityIdentifier
, responderPortIdentifier PortId
, serviceType           ApplicationIdentifier
, versionNumber         VersionNumber
, serviceInstanceIdentifier ServiceInstanceIdentifier
}
```

```
SleBindReturn          ::= SEQUENCE
{ performerCredentials  Credentials
, responderIdentifier    AuthorityIdentifier
, result                CHOICE
{ positive              [0] VersionNumber
, negative              [1] BindDiagnostic
}
}
```

```
SlePeerAbort           ::= PeerAbortDiagnostic
```

```
SleUnbindInvocation    ::= SEQUENCE
{ invokerCredentials    Credentials
, unbindReason          UnbindReason
}
```

```
SleUnbindReturn        ::= SEQUENCE
{ responderCredentials  Credentials
, result               CHOICE
{ positive             [0] NULL
}
}
```

```

-- =====
-- The second part of the module definition contains the types
-- used by the SLE-PDUs declared in the first part.
-- =====

ApplicationIdentifier ::= INTEGER
{
  rtnAllFrames          (0)
, rtnInsert             (1)
, rtnChFrames           (2)
  -- rtnChFrames includes rtnMcFrames and rtnVcFrames
, rtnChFsh              (3)
  -- rtnChFsh includes rtnMcFsh and rtnVcFsh
, rtnChOcf              (4)
  -- rtnChOcf includes rtnMcOcf and rtnVcOcf
, rtnBitstr             (5) -- AOS
, rtnSpacePkt           (6)
, fwdAosSpacePkt        (7)
, fwdAosVca             (8)
, fwdBitstr             (9)
, fwdProtoVcdu          (10)
, fwdInsert             (11)
, fwdCVcdu              (12)
, fwdTcSpacePkt         (13) -- conventional telecommand
, fwdTcVca              (14) -- conventional telecommand
, fwdTcFrame            (15)
, fwdCltu               (16)
}

ATTRIBUTE ::= CLASS
{
  &attributeId          OBJECT IDENTIFIER UNIQUE
, &AttributeType
}

AttributeTypeAndValue ::= SEQUENCE
{
  attributeIdentifier    ATTRIBUTE.&attributeId

  ({ServManagementAttributeSet})
, attributeValue        ATTRIBUTE.&AttributeType
  ({ServManagementAttributeSet}
  {@attributeIdentifier})
}

AuthorityIdentifier ::= VisibleString (SIZE (3 .. 16))

BindDiagnostic ::= INTEGER
{
  accessDenied          (0)
, serviceTypeNotSupported (1)
, versionNotSupported   (2)
, noSuchServiceInstance (3)
, alreadyBound          (4)
, siNotAccessibleToThisInitiator (5)
, inconsistentServiceType (6)
, invalidTime           (7)
, outOfService          (8)
, otherReason           (127)
}

LogicalPortName ::= VisibleString (SIZE (1 .. 128))

```

```
PeerAbortDiagnostic ::= INTEGER
{
  accessDenied          (0)
, unexpectedResponderId (1)
, operationalRequirement (2)
, protocolError         (3)
, communicationsFailure  (4)
, encodingError          (5)
, returnTimeout         (6)
, endOfServiceProvisionPeriod (7)
, unsolicitedInvokeId    (8)
, otherReason           (127)
}
-- The range of this INTEGER shall be (0 .. 255).
-- (128 .. 255) is reserved for diagnostic codes
-- that are specific to the communications technology used.
}

PortId ::= LogicalPortName

RDNSequence ::= SEQUENCE OF RelativeDistinguishedName

RelativeDistinguishedName ::= SET SIZE (1) OF AttributeTypeAndValue

ServiceInstanceIdentifier ::= RDNSequence

ServManagementAttributeSet ATTRIBUTE ::= {...}
-- For a real implementation, the above type definition must
-- be replaced by importing the corresponding information object set
-- specification from the applicable service management specification.
-- It is assumed that all attributes are of the type VisibleString
-- See also [F4].

UnbindReason ::= INTEGER
{
  end          (0)
, suspend      (1)
, versionNotSupported (2)
, other        (127)
}

VersionNumber ::= IntPosShort

END
```

A.2.3 SLE TRANSFER SERVICE COMMON PDUS

CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS    ReportingCycle
,          SleAcknowledgement
,          SleScheduleStatusReportInvocation
,          SleScheduleStatusReportReturn
,          SleStopInvocation
;

IMPORTS    Credentials
,          Diagnostics
,          InvokeId
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES
;

```

```

-- =====
-- The first part of the module definition contains the SLE-PDUS
-- =====

```

```

SleAcknowledgement ::= SEQUENCE
{
  credentials      Credentials
, invokeId         InvokeId
, result           CHOICE
  {
    positiveResult [0] NULL
  , negativeResult [1] Diagnostics
  }
}

SleScheduleStatusReportInvocation ::= SEQUENCE
{
  invokerCredentials Credentials
, invokeId           InvokeId
, reportRequestType  ReportRequestType
}

SleScheduleStatusReportReturn ::= SEQUENCE
{
  performerCredentials Credentials
, invokeId             InvokeId
, result               CHOICE
  {
    positiveResult [0] NULL
  , negativeResult [1] DiagnosticScheduleStatusReport
  }
}

SleStopInvocation ::= SEQUENCE
{
  invokerCredentials Credentials
, invokeId           InvokeId
}

```

```
-- =====
-- The second part of the module definition contains the types
-- used by the SLE-PDUs declared in the first part.
-- =====

DiagnosticScheduleStatusReport ::= CHOICE
{ common          [0] Diagnostics
, specific        [1] INTEGER
  { notSupportedInThisDeliveryMode (0) -- not used for
                                     -- version 1 forward transfer
                                     -- services
    , alreadyStopped (1)
    , invalidReportingCycle (2)
  }
}

-- The cycle duration is expressed in seconds
ReportingCycle ::= INTEGER (2 .. 600)

ReportRequestType ::= CHOICE
{ immediately [0] NULL
, periodically [1] ReportingCycle
, stop [2] NULL
}

END
```

A.2.4 CLTU TRANSFER STRUCTURES

CCSDS-SLE-TRANSFER-SERVICE-CLTU-STRUCTURES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS  BufferSize
,        CltuData
,        CltuGetParameter
,        CltuIdentification
,        CltuLastProcessed
,        CltuLastOk
,        CltuNotification
,        CltuParameterName
,        CltuStatus
,        ConditionalCltuIdentification
,        DiagnosticCltuGetParameter
,        DiagnosticCltuStart
,        DiagnosticCltuThrowEvent
,        DiagnosticCltuTransferData
,        EventThrownId
,        NumberOfCltusProcessed
,        NumberOfCltusRadiated
,        NumberOfCltusReceived
,        ProductionStatus
,        UplinkStatus
;
IMPORTS  ConditionalTime
,        DeliveryMode
,        Diagnostics
,        ForwardDuStatus
,        IntPosLong
,        IntPosShort
,        IntUnsignedLong
,        ParameterName
,        SpaceLinkDataUnit
,        Time
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

        ReportingCycle
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS
;

-- Available buffer in octets
BufferSize                ::= IntUnsignedLong

CltuData                  ::= SpaceLinkDataUnit

```

```

CltuDeliveryMode          ::= DeliveryMode
( fwdOnline
)

CltuGetParameter          ::= CHOICE
{ parBitLockRequired      [0] SEQUENCE
  { parameterName          ParameterName (bitLockRequired)
  , parameterValue          INTEGER
    { yes                    (0)
    , no                     (1)
    }
  }
, parCltuIdentification   [1] SEQUENCE
  { parameterName          ParameterName (expectedSlduIdentification)
  , parameterValue          CltuIdentification
  }
, parEventInvocationIdentification [2] SEQUENCE
  { parameterName          ParameterName
    (expectedEventInvocationIdentification)
  , parameterValue          EventThrownId
  }
, parSubcarrierToBitRateRatio [3] SEQUENCE
  { parameterName          ParameterName (subcarrierToBitRateRatio)
  , parameterValue          SubcarrierDivisor
  }
, parDeliveryMode          [4] SEQUENCE
  { parameterName          ParameterName (deliveryMode)
  , parameterValue          CltuDeliveryMode
  }
, parMaximumCltuLength     [5] SEQUENCE
  { parameterName          ParameterName (maximumSlduLength)
  , parameterValue          INTEGER (96 .. 32784)
  }
, parModulationFrequency   [6] SEQUENCE
  { parameterName          ParameterName (modulationFrequency)
  , parameterValue          ModulationFrequency
  }
, parModulationIndex       [7] SEQUENCE
  { parameterName          ParameterName (modulationIndex)
  , parameterValue          ModulationIndex
  }
, parPlopInEffect          [8] SEQUENCE
  { parameterName          ParameterName (plopInEffect)
  , parameterValue          INTEGER
    { plop1                  (0)
    , plop2                  (1)
    }
  }
, parReportingCycle        [9] SEQUENCE
  { parameterName          ParameterName (reportingCycle)
  , parameterValue          CurrentReportingCycle
  }
, parReturnTimeout         [10] SEQUENCE

```

```

    { parameterName      ParameterName (returnTimeoutPeriod)
      , parameterValue   TimeoutPeriod
    }
  , parRfAvailableRequired [11] SEQUENCE
    { parameterName      ParameterName (rfAvailableRequired)
      , parameterValue   INTEGER
        { yes            (0)
          , no            (1)
        }
    }
}

```

```

CltuIdentification ::= IntUnsignedLong

```

```

CltuLastOk ::= CHOICE
{ noCltuOk      [0] NULL
, cltuOk        [1] SEQUENCE
  { cltuIdentification CltuIdentification
    , radiationStopTime Time
  }
}

```

```

CltuLastProcessed ::= CHOICE
{ noCltuProcessed [0] NULL
, cltuProcessed   [1] SEQUENCE
  { cltuIdentification CltuIdentification
    , radiationStartTime ConditionalTime
    , cltuStatus        CltuStatus
  }
}

```

```

CltuNotification ::= CHOICE
{ cltuRadiated      [0] NULL
, slduExpired        [1] NULL
, unableToProcess    [2] NULL
, productionHalted   [3] NULL
, productionOperational [4] NULL
, bufferEmpty        [5] NULL
, actionListCompleted [6] EventThrownId
, actionListNotCompleted [7] EventThrownId
, eventConditionEvFalse [8] EventThrownId
}

```

```

CltuParameterName ::= ParameterName
( bitLockRequired
| deliveryMode
| expectedSlduIdentification
| expectedEventInvocationIdentification
| maximumSlduLength
| modulationFrequency
| modulationIndex
| plopInEffect
| reportingCycle
| returnTimeoutPeriod

```

```

| rfAvailableRequired
| subcarrierToBitRateRatio
)

CltuStatus ::= ForwardDuStatus
(
| radiated
| expired
| interrupted
| productionStarted
| undefined
)

ConditionalCltuIdentification ::= CHOICE
{
noCltuProcessed      [0] NULL
, cltuProcessed      [1] CltuIdentification
}

CurrentReportingCycle ::= CHOICE
{
periodicReportingOff [0] NULL
, periodicReportingOn [1] ReportingCycle
}

DiagnosticCltuGetParameter ::= CHOICE
{
common      [0] Diagnostics
, specific  [1] INTEGER
  {
unknownParameter (0)
}
}

DiagnosticCltuStart ::= CHOICE
{
common      [0] Diagnostics
, specific  [1] INTEGER
  {
outOfService (0)
, unableToComply (1)
, productionTimeExpired (2)
}
}

DiagnosticCltuThrowEvent ::= CHOICE
{
common      [0] Diagnostics
, specific  [1] INTEGER
  {
operationNotSupported (0)
, eventInvocIdOutOfSequence (1)
, noSuchEvent (2)
}
}

DiagnosticCltuTransferData ::= CHOICE
{
common      [0] Diagnostics
, specific  [1] INTEGER
  {
unableToProcess (0)
, unableToStore (1)
, outOfSequence (2)
, inconsistentTimeRange (3)
}
}

```

```
, invalidTime           (4)
, lateSldu              (5)
, invalidDelayTime      (6)
, cltuError             (7)
}
}

EventThrownId           ::= IntUnsignedLong

-- either the subcarrier frequency, or,
-- in case of direct modulation, the bit rate, measured in 1/10 Hz
ModulationFrequency     ::= IntPosLong

-- expressed in carrier suppression in 1/100 dB
ModulationIndex         ::= IntPosShort

NumberOfCltusProcessed  ::= IntUnsignedLong

NumberOfCltusRadiated   ::= IntUnsignedLong

NumberOfCltusReceived   ::= IntUnsignedLong

ProductionStatus        ::= INTEGER
{ operational           (0)
, configured            (1)
, interrupted           (2)
, halted                (3)
}

-- Divisor of the subcarrier frequency
-- If direct carrier modulation, the value is 1
SubcarrierDivisor       ::= IntPosShort

-- measured in seconds
TimeoutPeriod           ::= INTEGER (1 .. 600)

UplinkStatus            ::= INTEGER
{ uplinkStatusNotAvailable (0)
, noRfAvailable           (1)
, noBitLock                (2)
, nominal                  (3)
}

END
```

A.2.5 CLTU INCOMING PDUS**CCSDS-SLE-TRANSFER-SERVICE-CLTU-INCOMING-PDUS****DEFINITIONS****IMPLICIT TAGS**

```
::= BEGIN
```

```
IMPORTS    ConditionalTime
,          Credentials
,          Duration
,          IntPosShort
,          InvokeId
,          SlduStatusNotification
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

          SleScheduleStatusReportInvocation
,          SleStopInvocation
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

          CltuData
,          CltuIdentification
,          CltuParameterName
,          ConditionalCltuIdentification
,          EventThrownId
FROM CCSDS-SLE-TRANSFER-SERVICE-CLTU-STRUCTURES

          SleBindInvocation
,          SlePeerAbort
,          SleUnbindInvocation
FROM CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES
;
```

```
-- =====
-- The first part of the module definition contains the CLTU type
-- that contains all the possible PDUs the provider may receive.
-- =====
```

```
CltuUserToProviderPdu ::= CHOICE

{ cltuBindInvocation           [100]  SleBindInvocation
, cltuUnbindInvocation         [102]  SleUnbindInvocation
, cltuStartInvocation          [0]    CltuStartInvocation
, cltuStopInvocation           [1]    SleStopInvocation
, cltuTransferDataInvocation   [2]    CltuTransferDataInvocation
, cltuScheduleStatusReportInvocation [3]
                                SleScheduleStatusReportInvocation
```

```

, cltuGetParameterInvocation      [4]    CltuGetParameterInvocation
, cltuThrowEventInvocation        [5]    CltuThrowEventInvocation
, cltuPeerAbortInvocation         [104]   SlePeerAbort
}

```

```

-- =====
-- The second part of the module definition contains the types
-- used by the RAF-PDUs declared in the first part.
-- =====

```

```

CltuGetParameterInvocation ::= SEQUENCE
{
  invokerCredentials      Credentials
, invokeId                InvokeId
, cltuParameter           CltuParameterName
}

```

```

CltuStartInvocation        ::= SEQUENCE
{
  invokerCredentials      Credentials
, invokeId                InvokeId
, firstCltuIdentification ConditionalCltuIdentification
}

```

```

CltuThrowEventInvocation   ::= SEQUENCE
{
  invokerCredentials      Credentials
, invokeId                InvokeId
, eventIdIdentification    IntPosShort
, eventThrownId           EventThrownId
}

```

```

CltuTransferDataInvocation ::= SEQUENCE
{
  invokerCredentials      Credentials
, invokeId                InvokeId
, cltuIdentification      CltuIdentification
, earliestTransmissionTime ConditionalTime
, latestTransmissionTime  ConditionalTime
, delayTime               Duration
, slduRadiationNotification SlduStatusNotification
, cltuData                CltuData
}

```

```

END

```

A.2.6 CLTU OUTGOING PDUS

CCSDS-SLE-TRANSFER-SERVICE-CLTU-OUTGOING-PDUS

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

IMPORTS      ConditionalTime,
              Credentials,
              InvokeId
,            Time
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

              SleAcknowledgement
,            SleScheduleStatusReportReturn
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

              SleBindReturn
,            SleUnbindReturn
,            SlePeerAbort
FROM CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES

              BufferSize
,            CltuGetParameter
,            CltuIdentification
,            CltuLastProcessed
,            CltuLastOk
,            CltuNotification
,            CltuStatus
,            ConditionalCltuIdentification
,            DiagnosticCltuGetParameter
,            DiagnosticCltuStart
,            DiagnosticCltuThrowEvent
,            DiagnosticCltuTransferData
,            EventThrownId
,            NumberOfCltusProcessed
,            NumberOfCltusRadiated
,            NumberOfCltusReceived
,            ProductionStatus
,            UplinkStatus
FROM CCSDS-SLE-TRANSFER-SERVICE-CLTU-STRUCTURES
;

```

```

-- =====
-- The first part of the module definition contains the CLTU type
-- that contains all the possible PDUs the provider may send.
-- =====

```

```

CltuProviderToUserPdu      ::= CHOICE
{ cltuBindReturn            [101]  SleBindReturn
, cltuUnbindReturn          [103]  SleUnbindReturn

```

```

, cltuStartReturn          [0]    CltuStartReturn
, cltuStopReturn           [1]    SleAcknowledgement
, cltuTransferDataReturn   [2]    CltuTransferDataReturn
, cltuAsyncNotifyInvocation [3]    CltuAsyncNotifyInvocation
, cltuScheduleStatusReportReturn [4]    SleScheduleStatusReportReturn
, cltuStatusReportInvocation [5]    CltuStatusReportInvocation
, cltuGetParameterReturn   [7]    CltuGetParameterReturn
, cltuThrowEventReturn     [8]    CltuThrowEventReturn
, cltuPeerAbortInvocation  [104]   SlePeerAbort
}

```

```

-- =====
-- The second part of the module definition contains the types
-- used by the RAF-PDUs declared in the first part.
-- =====

```

```

CltuAsyncNotifyInvocation ::= SEQUENCE
{
  invokerCredentials      Credentials
, cltuNotification        CltuNotification
, cltuLastProcessed       CltuLastProcessed
, cltuLastOk              CltuLastOk
, productionStatus        ProductionStatus
, uplinkStatus            UplinkStatus
}

```

```

CltuGetParameterReturn ::= SEQUENCE
{
  performerCredentials      Credentials
, invokeId                  InvokeId
, result                    CHOICE
  {
    positiveResult          [0] CltuGetParameter
  , negativeResult          [1] DiagnosticCltuGetParameter
  }
}

```

```

CltuStartReturn ::= SEQUENCE
{
  performerCredentials      Credentials
, invokeId                  InvokeId
, result                    CHOICE
  {
    positiveResult          [0] SEQUENCE
      {
        radiationStartTime    Time
      , radiationStopTime     ConditionalTime
      }
  , negativeResult          [1] DiagnosticCltuStart
  }
}

```

```

}
```

```

CltuStatusReportInvocation ::= SEQUENCE
{
  invokerCredentials      Credentials
, cltuLastProcessed      CltuLastProcessed
, cltuLastOk              CltuLastOk
, cltuProductionStatus    ProductionStatus
, uplinkStatus            UplinkStatus
, numberOfCltusReceived   NumberOfCltusReceived
, numberOfCltusProcessed  NumberOfCltusProcessed
, numberOfCltusRadiated   NumberOfCltusRadiated
, cltuBufferAvailable     BufferSize
}

```

```

CltuThrowEventReturn ::= SEQUENCE
{
  performerCredentials    Credentials
, invokeId                InvokeId
, eventThrownId           EventThrownId
, result                  CHOICE
  {
    positiveResult        [0] NULL
    , negativeResult       [1] DiagnosticCltuThrowEvent
  }
}

```

```

CltuTransferDataReturn ::= SEQUENCE
{
  performerCredentials    Credentials
, invokeId                InvokeId
, cltuIdentification      CltuIdentification
, cltuBufferAvailable     BufferSize
, result                  CHOICE
  {
    positiveResult        [0] NULL
    , negativeResult       [1] DiagnosticCltuTransferData
  }
}

```

```

END

```

ANNEX B**INDEX TO DEFINITIONS**

(This annex **is not** part of the Recommendation)

This annex provides an index to the terminology defined in the references.

Term	Reference
(data) type	reference [6]
(data) value	reference [6]
abstract binding	reference [1]
abstract object	reference [1]
abstract port	reference [1]
abstract service	reference [1]
abstract syntax	reference [7]
Abstract Syntax Notation One (ASN.1)	reference [6]
acquisition sequence	reference [2]
application layer	reference [7]
application process	reference [7]
association	subsection 1.6.1.8.1
Carrier Modulation Mode (CMM)	reference [2]
CLTU channel	reference [1]
Command Link Control Word (CLCW)	reference [3]
Command Link Transmission Unit (CLTU)	reference [2]
communications service	subsection 1.6.1.8.2
concatenation	reference [7]
confirmed operation	subsection 1.6.1.8.3

distinguished name (DN)	reference [9]
expedited (BD) service	reference [3]
flow control	reference [7]
Forward CLTU service	reference [1]
idle sequence	reference [2]
initiator	reference [8]
invocation	subsection 1.6.1.8.4
invoker	reference [1]
Mission Data Operation System (MDOS)	reference [1]
Mission User Entity (MUE)	reference [1]
object identifier	reference [6]
offline delivery mode	reference [1]
online delivery mode	reference [1]
Open System Interconnection (OSI)	reference [7]
operation	reference [1]
parameter	subsection 1.6.1.8.5
performance	subsection 1.6.1.8.6
performer	reference [1]
physical channel	reference [1]
Physical Layer Operations Procedure (PLOP)	reference [2]
port identifier	subsection 1.6.1.8.7
real system	reference [7]
relative distinguished name (RDN)	reference [9]
responder	reference [8]

return	subsection 1.6.1.8.8
sequence controlled (AD) service	reference [3]
service access point (SAP)	reference [7]
service instance provision period	subsection 1.6.1.8.9
service provider (provider)	reference [1]
service user (user)	reference [1]
SLE Complex	reference [1]
SLE Complex Management	reference [1]
SLE data channel	reference [1]
SLE functional group (SLE-FG)	reference [1]
SLE protocol data unit (SLE-PDU)	reference [1]
SLE service data unit (SLE-SDU)	reference [1]
SLE service package	reference [1]
SLE System	reference [1]
SLE transfer service instance	reference [1]
SLE transfer service production	reference [1]
SLE transfer service provision	reference [1]
SLE Utilization Management	reference [1]
space link	reference [1]
space link data channel	reference [1]
space link data unit (SL-DU)	reference [1]
space link session	reference [1]
unconfirmed operation	subsection 1.6.1.8.10

ANNEX C**MAPPING OF SLE-PDUS TO SLE-SDUS**

(This annex **is not** part of the Recommendation.)

Part of the data exchanged between an SLE user, i.e. a Mission User Entity (MUE) within an Mission Data Operation System (MDOS), and a Forward CLTU service provider within an SLE Complex, is directly related to the Space Link Data Units (SL-DU) exchanged with the spacecraft or to the status of the channel carrying the SL-DUs. The SLE Reference Model (reference [1]) refers to these data units as SLE Service Data Units (SLE-SDU).

In a real operational environment, the MDOS and the SLE Complex must exchange these SLE-SDUs via a remote interface, which necessitates additional data. The data units embedding the SLE-SDUs for the purpose of exchanging them across a remote interface or carrying data solely for the “administration” of the remote interface are referred to as SLE Protocol Data Units (SLE-PDUs). Examples of SLE-PDUs carrying solely administrative information are the SLE-PDUs representing the CLTU-BIND, CLTU-UNBIND, CLTU-START and CLTU-STOP operations, while the SLE-SDUs are mostly embedded in the CLTU-TRANSFER-DATA SLE-PDU.

The SLE-SDU is a formalism of the SLE Reference Model (reference [1]). In a real system, SLE-SDUs do not exist as separable, identifiable data units. They are documented in Table C-1 to provide traceability to the SLE Reference Model.

Table C-1 lists the pseudo CLTU SLE-SDUs that would exist if the SLE-SDUs were indeed realized. For each of these pseudo SLE-SDUs, the corresponding CLTU SLE-PDU carrying a Forward CLTU operation invocation or return is identified, and the parameters of that operation that would constitute the content of the pseudo SLE SDU.

Table C-1: Mapping of Pseudo-CLTU SLE-SDUs to CLTU SLE-PDUs

Pseudo CLTU SLE-SDU	Corresponding CLTU SLE-PDU	Operation Parameters
CLTU-TRANSFER-DATA-INVOCATION	CLTU-TRANSFER-DATA invocation	cltu-identification earliest-radiation-time latest-radiation-time delay-time report data
CLTU-TRANSFER-DATA-RETURN	CLTU-TRANSFER-DATA return	result diagnostic: All values except 'duplicate Invoke-ID' and 'unable to store'.
CLTU-ASYNC-NOTIFY-INVOCATION	CLTU-ASYNC-NOTIFY invocation	notification-type: All values except 'action list completed', 'action list not completed', and 'event condition evaluated to false'. cltu-identification cltu-status radiation-start-time radiation-stop-time production-status

ANNEX D**ACRONYMS**

(This annex **is not** part of the Recommendation)

This annex expands the acronyms used throughout this Recommendation.

ASDC	Abstract Service Definition Conventions
ASN.1	Abstract Syntax Notation One
C	Conditional
CCSDS	Consultative Committee for Space Data Systems
CLCW	Command Link Control Word
CLTU	Command Link Transmission Unit
CLTU-	Forward CLTU service (operation prefix)
CM-P	Complex Management Port
COP	Command Operation Procedure
FG	Functional Group
FWD	Forward
IP	Internet Protocol
ISO	International Organization for Standardization
M	Mandatory
MDOS	Mission Data Operation System
MUE	Mission User Entity
OCF	Operational Control Field
PDU	Protocol Data Unit
RF	Radio Frequency
SAP	Service Access Point
SDU	Service Data Unit
SLE	Space Link Extension
SLE-FG	SLE Functional Group
SLE-PDU	SLE Protocol Data Unit
SLE-SDU	SLE Service Data Unit
SL-DU	Space Link Data Unit

CCSDS HISTORICAL DOCUMENT

DRAFT CCSDS RECOMMENDATION FOR SLE FORWARD CLTU SERVICE

TC	Telecommand
TCP	Transmission Control Protocol
T-P	Transfer Provider Port
T-U	Transfer User Port
UTC	Universal Time Coordinated

ANNEX E

CONFORMANCE OPTIONS MATRIX

(This annex is part of the Recommendation)

This annex provides the Conformance Matrix for implementations of Space Link Extension (SLE) Forward Command Link Transmission Unit (CLTU) Service systems. An SLE Forward CLTU Service system will be considered to be 'conformant' if the mandatory elements identified in the matrix are implemented as described in this Recommendation.

Service Operations and their Parameters	Optional / Mandatory
CLTU-BIND CLTU-UNBIND	These operations are mandatory. All parameters are also mandatory with the exception of: <code>initiator-credentials</code> <code>responder-credential</code> Support of the <code>version-number</code> negotiation as part of the CLTU-BIND operation is optional.
CLTU-START CLTU-STOP CLTU-TRANSFER-DATA CLTU-ASYNC-NOTIFY CLTU-SCHEDULE-STATUS-REPORT CLTU-STATUS-REPORT CLTU-GET-PARAMETER	These operations are mandatory. All parameters are also mandatory with the exception of: <code>invoker-credentials</code> <code>performer-credentials</code> <code>stop-production-time</code>
CLTU-THROW-EVENT	This operation is optional.
CLTU-PEER-ABORT	This operation is mandatory. The <code>diagnostic</code> parameter is also mandatory.

Other Sections	Optional / Mandatory
Provider Buffering Requirements (see 3.1.11)	Mandatory minimum size is 1024 maximum sized CLTUs
State transition table (see 4.2)	Mandatory
ASN.1 Protocol Specification (ANNEX A)	All data types are mandatory. Implementations may vary as described in the annex.

ANNEX F

INFORMATIVE REFERENCES

(This annex **is not** part of the Recommendation)

- [F1] *Procedures Manual for the Consultative Committee for Space Data Systems*. CCSDS A00.0-Y-7. Yellow Book. Issue 7. Washington, D.C.: CCSDS, November 1996.
- [F2] *Telecommand Summary of Concept and Rationale*. Report Concerning Space Data System Standards, CCSDS 200.0-G-6. Green Book. Issue 6. Washington, D.C.: CCSDS, January 1987.
- [F3] *Standard Terminology, Conventions, and Methodology (TCM) for Defining Data Services*. Report Concerning Space Data Systems Standards, CCSDS 910.2-G-1. Green Book. Issue 1. Washington, D.C.: CCSDS, November 1994.
- [F4] *Space Link Extension Service Management Specification*. Draft Recommendation for Space Data System Standards, CCSDS 910.5-W-1.18. White Book. Issue 1.18. CCSDS, 1998.
- [F5] *Cross Support Concept – Part 1: Space Link Extension Services*; Report Concerning Space Data Systems Standards. CCSDS 910.3-G-1. Green Book. Issue 1. Washington, DC: CCSDS, May 1995.